

# Outline (until the next coffee break)

- Lecture: Modeling of regulatory activity with sequence models
- Hands-on: Training a sequence-based model
- Discussion of pre-training and fine-tuning approaches
- Hands-on: Interpreting models with in-silico mutagenesis



# Sequence modeling

**ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)**

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Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH  
July 20, 2025



UNIVERSITÄT ZU LÜBECK



UNIVERSITÄTSKLINIKUM  
Schleswig-Holstein

**BIH** Berlin Institute  
of Health  
@Charité

# Scope of this session

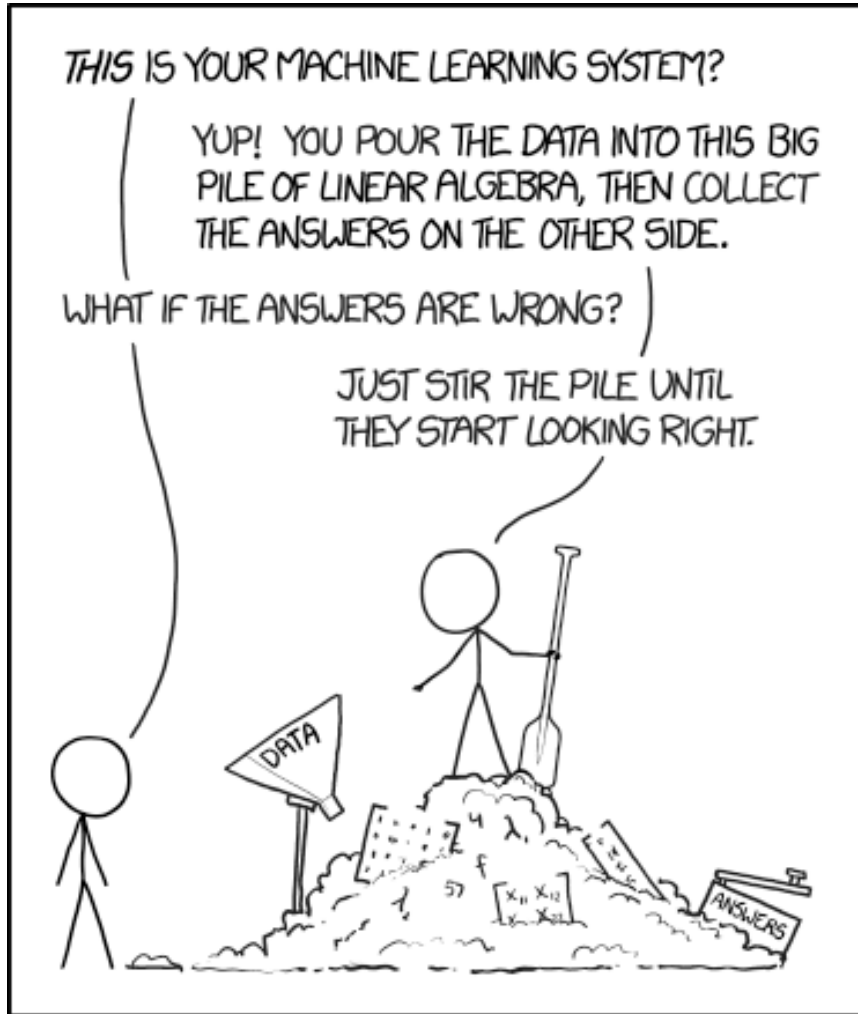
## Goals

- Intuitive understanding of machine learning and basic sequence models
- Neural networks (NNs): What are they? and How they work?
- Example: **DeepSTARR**
- Model Interpretation

## Questions we will cover:

- Why sequence-based modeling?
- How to leverage NNs alongside experimental data?
- What can we gain from NNs and how?

# Machine learning 101



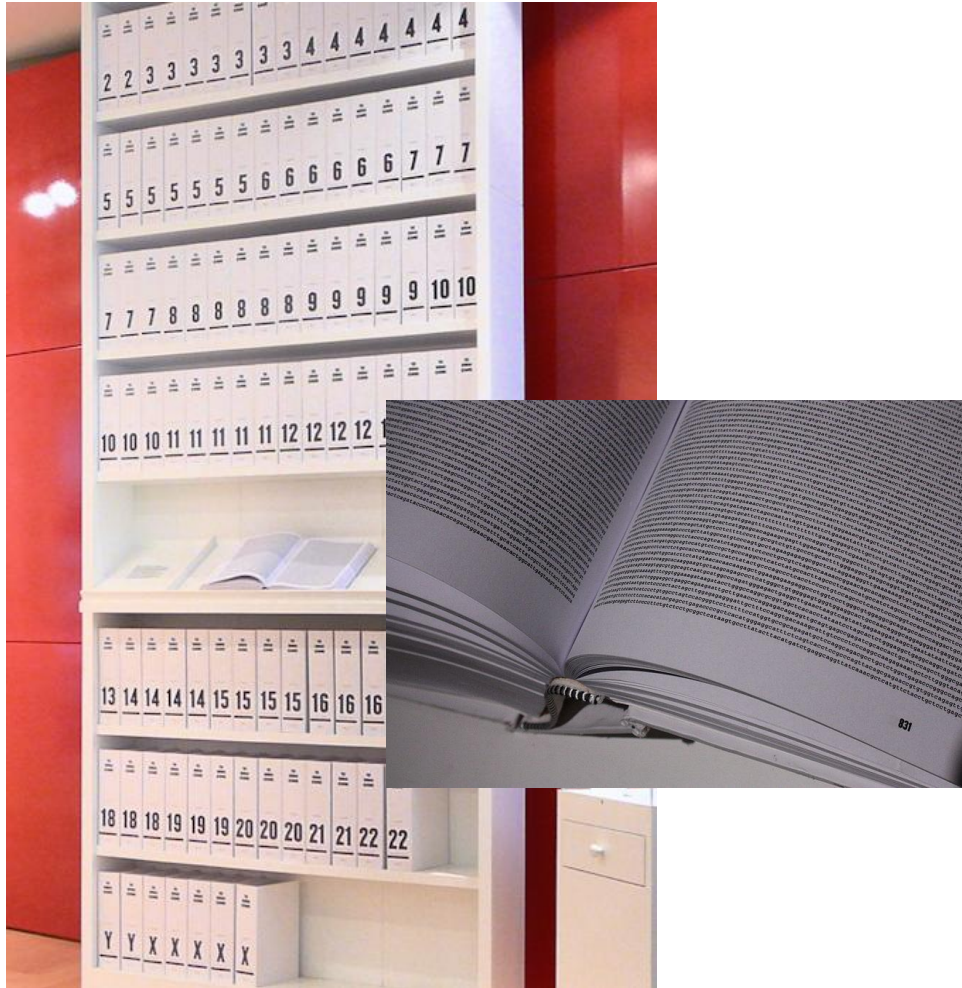
make computer systems ... learn  
generalizations ... (without explicit  
information) ... from patterns of data

<https://xkcd.com/1838/>

# Basic terminology

- **Data – typically transformed to derive features**
- **Typically, a data set (features + output/response) is split into:**
  - **train and test set**, entries within these sets are called **examples**
  - sometimes an (independent) **validation set** is used, and the **test set serves for optimizing hyperparameters** (i.e., all kinds of tuning parameters to the model, like the number of parameters or training iterations)
- **Responses can be categorical (classification) or numerical (regression)**
- **Model training is either:**
  - Supervised: response is available, and the model is trained to produce the response
  - Semisupervised: proportion of training examples does not have a response
  - Unsupervised: clustering (e.g. kmeans, graphs/cliques, self-organizing maps)
- **Model performance is assessed by comparing predicted responses with original responses**
- **Models can be trained on surrogate data / correlated problems, recently called “zero-shot”**

# Sequence modeling

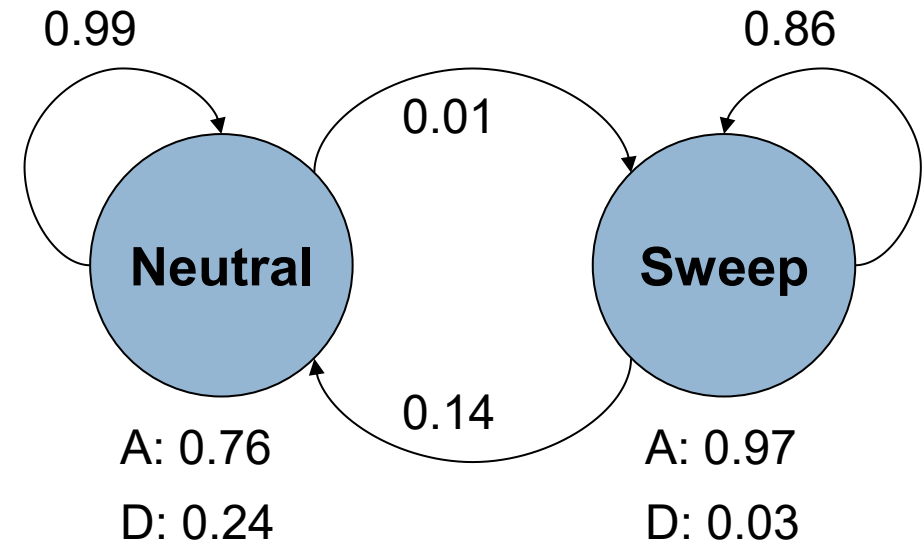


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TCTAACTCCACAGTATTTTTCCCCCTGGTGACTGACACAGCTGACCTAGAGCTAGAAGTCACC  
TCATCTGACCCCTGGTTGCTGGCTCTCCACTTGACAATGGAGAGGACATGGACCCTGAGGGC  
CTCTCCACACAGACTGGGCCTCACCTCTCCACAGAGGAATTCAGCTCAGGAGTAGATGTAC  
AGATGCCGGCACGATCTGCTTACAGCTTCTCATTGCAGGCATAGTCTTTTTTTTTTTTTTTTT  
TTTGGCTACTGAGCTCATCAAAGCTGGTGAGGACCCAGCTAAGGTTGTTTGCTGCTGCTGTG  
AGCACTTGAGTAGTAATTGGGGTGTGGTGAGGGAGAAAACAAATGCCCTCAAGGGGGCAGCCA  
CAGCCCCTAACTCTAGGCCTGACAGAGAAGGCAGGGAGCTCGGCAGACAGGTGCGTGGAGAC  
ATTCTGATCCCAATCTCACTTAAGGATTAAGGGAGGGGGCAGTAGGGAGACAGGATAGCCAG  
TCTTGTAATTTGAGACGGGCCCAGGATCCCTTATCCTGCAATCTGTCCTGCTGGACAGAGCA  
CTGTCCTCAGCACAGAGGCAAGGTCCTCAATAAAGAGCTGGGAGCAGTGACTGCAGACTCAG  
AGGGTCTCTTCTCCTGGATCCTTATCTGGCTTCCACCCCAGGAAGCCAGGCCATTGCTACAC  
CTGTGTGAGGCTCTCTAGGGAGATGGGGCTGGGGCTGGCCCTGCTTCCCTTCTGCATTTATC  
AAGTGCCTGTGACAGGGAAGACACCACAAGAGGGTCACCGACCTGAGGGCAGTGCAGTCCCA  
GGTCTTGAAATAGTGAAGTTCAGTTTCTAAACCCCTCTGTGCCTCCTTCTTGCTCTCTGTAA  
AATGCTGCGGTTGGAGAGATGGTGGATACAGGGCCTTTTAAAGTTGAATTTCCAGATGATTG  
GTTTCCAAGCTTTCTGAAAAATGAGGCCCTTATCAATGTTGAAAAATAGGTAAATCTCATT  
CTTTTTTATGACTGAACTTAAAGAATGTGATTGGATTGTTTGTAACCCAAAGTATAAAGGCT  
TTTTTTTTTTTTTCTGAGACAAGGTCTCACTCTGTACCTAGGCTGGATTGCAGTGGCATAA  
TCACAGCTCACTGCAGCCTTGACCTCCCAGGCTCAGGCGACCTTCCTGCCTCAGCCTCCTGA  
ATAGCTGGGATTATAGGTGTGCATCACTATGCCTAGTTAATTTTTGTATTTTTTGTAGAGAC  
GGGGCTTCATCATGTTGCCCAAATGCTTGATGTGCTTATTTACATTCCATGCCTGTATCAA  
AACATCTCATGTACCCCATACATGTATATACCTACTATGTATGCATAAAAATTTTTTAAAT  
AATTTTTAAATTAAAAAAATTCTGAGAATTTAAGCTTTTAAATCTCAGTTGAGTACTAAGTG  
AATGGTAATTAAAGACTAAAATTACTACCAGTTGTCATATTTAATTGAATTTAAATTTAC  
CATTGCAATCATAGAAGTCAACTCTTATTGAAAGCTTAGTCTGTGCCAAGCTCTATTCTAAA  
ATGTATTGACCCTAAATACATTTAGATAATGTATACATTATCTAAATGTAATAAAAAAATA  
ATAATTTTTGTTATTATTTCCATCCTATGGATGAAGCTGCTGAAACCCAAGTCCCACTACT  
GGGTGGTGGCAACACTATGAGCTGTTGGGAAAATATACTAAATCTATTATTAAGTTTAAGT  
AAAATAATTGAATATTATATAATTATATGAATTAATCAAATATAATTAAATATATTTCTATAA  
ATAAATATTATTAGAAATAATTCAAGGAGAAAAAAATCCCCAACCCAGAAAAACCTGAGG

Printed Human Genome of the Wellcome Collection

## A classic – Hidden Markov Models (HMMs)

- E.g. two state model of genomic regions under neutral evolution vs selection, based on sequence of single nucleotide polymorphisms in an ancestral (A) or derived (D) state (similar to “Occasionally dishonest casino” with fair vs loaded coin)
- HMMs can be used to:
  - Predict gene models, e.g. define Exon/Intron boundaries
  - Segment the genome based on sequence and/or sequence annotations (e.g. ChromHMM, Segway)



# kmer representation

```
CCTGGTGGAGAATTTAGTGACATTTGTTGGCTGCTTCTTAGGGAAGAAACCAA  
GTCTGACTCTAACTCCACAGTATTTTCCCCCTGGTGACTGACACAGCTGACCTAG  
AGCTAGAAGTCACCTCATCTGACCCCTGGTTGCTGGCTCTCCACTTGACAATGGA  
GAGGACATGGACCCTGAGGGCCTCTCCACACAGACTGGGCCTCACCTCTCCACA  
GAGGAATTCAGCTCAGGAGTAGATGTACAGATGCCGGCACGATCTGCTTACAGCT  
TCTCATTGCAGGCATAGTCTTTTTTTTTTTTTTTTTTGGCTACTGAGCTCATCA  
AAGCTGGTGAGGACCCAGCTAAGGTTGTTTGCTGCTGCTGTGAGCACTTGAGTAG  
TAATTGGGGTGTGGTGAGGGAGAAAACAAATGCCTCAAGGGGGCAGCCACAGCCC  
CTAACTCTAGGCCTGACAGAGAAGGCAGGGAGCTCGGCAGACAGGTGCGTGGAGA  
CATTCTGATCCCAATCTCACTTAAGGATTAAGGGAGGGGGCAGTAGGGAGACAGG  
ATAGCCAGTCTTGTAATTGAGACGGGCCAGGATCCCTTATCCTGCAATCTGTC  
CTGCTGGACAGAGCACTGTCCTCAGCACAGAGGCAAGGTCCTCAATAAAGAGCTG  
GGAGCAGTGACTGCAGACTCAGAGGGTCTCTTCTCCTGGATCCTTATCTGGCTTC  
CACCCCAGGAAGCCAGGCCATTGCTACACCTGTGTGAGGCTCTCTAGGGAGATGG  
GGCTGGGGCTGGCCCTGCTTCCCTTCTGCATTTATCAAGTGCCTGTGACAGGGAA  
GACACCACAAGAGGGTCACCGACCTGAGGGCAGTGCAGTCCCAGGTCTTGAAATA  
GTGAAGTTTCAGTTTCTAAACCCCTCTGTGCCTCCTTCTTGCTCTCTGTAAAATGC  
TGCGGTTGGAGAGATGGTGGATACAGGGCCTTTTAAGTTGAATTTCCAGATGAT  
TGGTTTCCAAGCTTTCTGAAAAATGAGGCCCTTATCAATGTTGAAAAATAGGTA  
AATCTCATTCTTTTTTATGACTGAACTTAAAGAATGTGATTGGATTGTTGTAAAC  
CCAAAGTATAAAGGCTTTTTTTTTTTTTTCTGAGACAAGGTCTCACTCTGTCACC  
TAGGCTGGATTGCAGTGGCATAATCACAGCTCACTGCAGCCTTGACCTCCCAGGC  
TCAGGCGACCTTCCTGCCTCAGCCTCCTGAATAGCTGGGATTATAGGTGTGCATC  
ACTATGCCTAGTTAATTTTTTGTATTTTTTGTAGAGACGGGGCTTCATCATGTTGC  
CCAAATGCTTGATGTGCTTATTTTACATTCATGCCTGTATCAAAACATCTCATG  
TACCCCATACATGTATATACCTACTATGTATGCATAAAAAATTTTTTAAATAATT  
TTTAAAATTAATAAATTTCTGAGAATTTAAGCTTTTAATCTCAGTTGAGTACTAA  
GTGAATGGTAATTAAAGACTAAAAATTACTACCAGTTGTCATATTTTAATTGAATT  
TAAAAATTTACCATTGCAATCATAGAAGTCAACTCTTATTGAAAGCTTAGTCTGTG  
CCAAGCTCTATTCTAAAATGTATTGACCCATAAATACATTTAGATAATGTATACAT  
TATCTAAAATGTAATAAAAAATAATAATTTTTGTTATTATTTCCATCCTATGGA  
TGAAGCTGCTGAAACCCAAGGTCCCACTACTGGGTGGTGGCAACACTATGAGCTG  
TTGGGAAAATATACTAAATCTATTATTAAGTTTAAGTAAATAAATTGAATATTA  
TATAATTATATGAATTAATCAAATATAATTAAATATATTTTCAATAAATAATATTA  
TTAGAAATAATTCAAGGAGAAAAAAATCCCCAACCCAGAAAAACCTGAGG
```

substrings/kmers  
of length 8



```
'TTTTTTTTT', 19  
'AAAAAAAT', 3  
'AAATAATT', 3  
'TAATTTTTT', 3  
'TATAATTA', 2  
'ATATAATT', 2  
'AATATTAT', 2  
'TATTATTA', 2  
'AAATATAA', 2  
'AAAATAAT', 2  
'AAAATGTA', 2  
'TAAAATGT', 2  
'CTAAAATG', 2  
'TCTAAAAT', 2  
'AATTGAAT', 2  
'TAATTGAA', 2  
'TAATTAAA', 2  
...
```

Identify what is

- unique
- frequent
- differentially frequent

...



# gapped-kmer SVMs

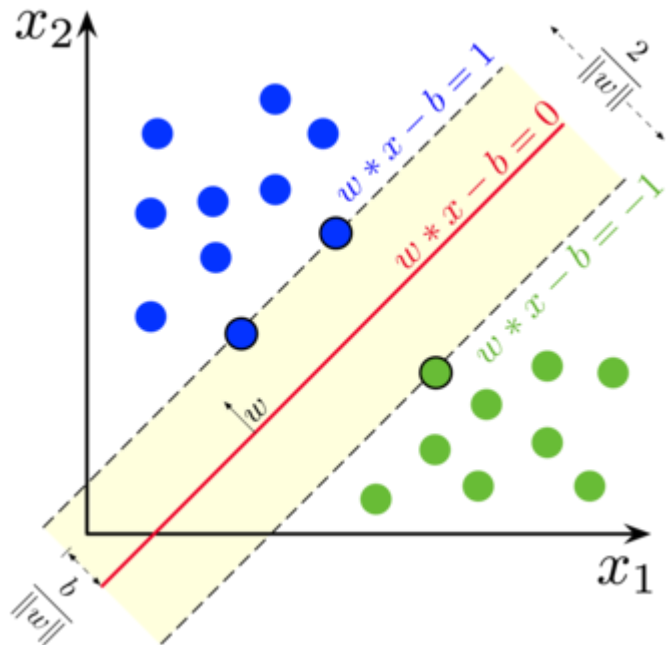
Gaps as „no information“ positions

GGAA TTGTAG

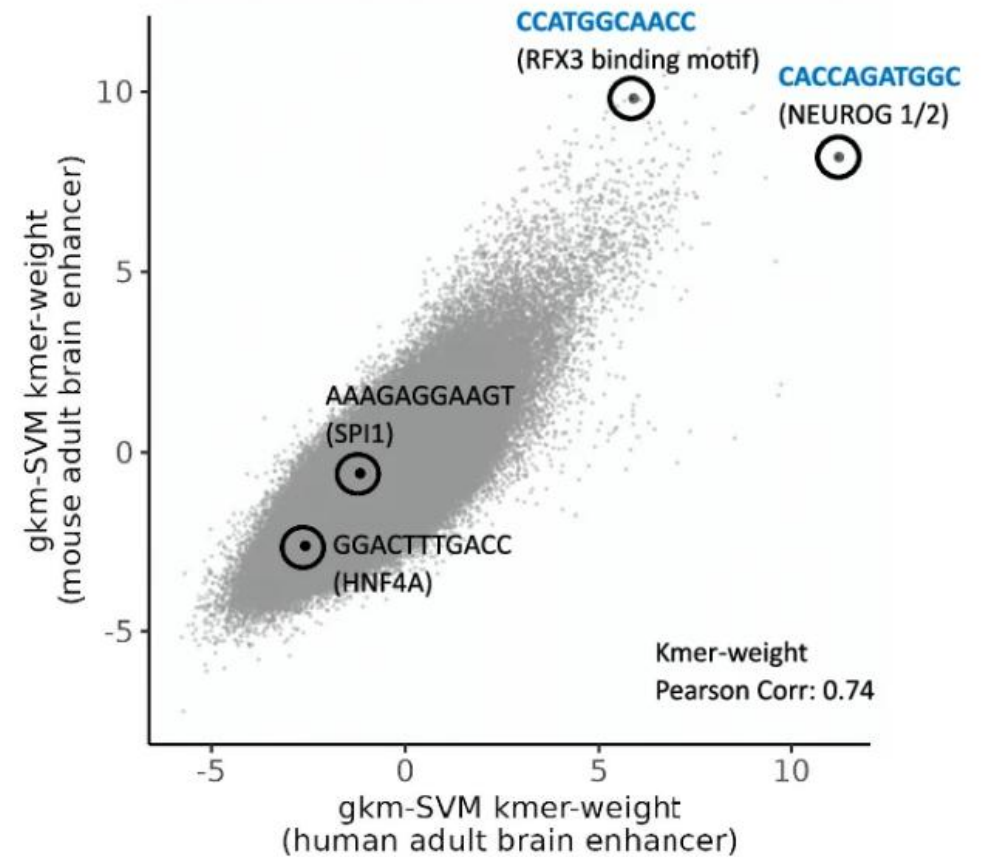
-G--TTGTAG

CGGT TTGTAG

-G--TTGTAG



Regulatory vocabularies  
of gkm-SVM brain enhancer models



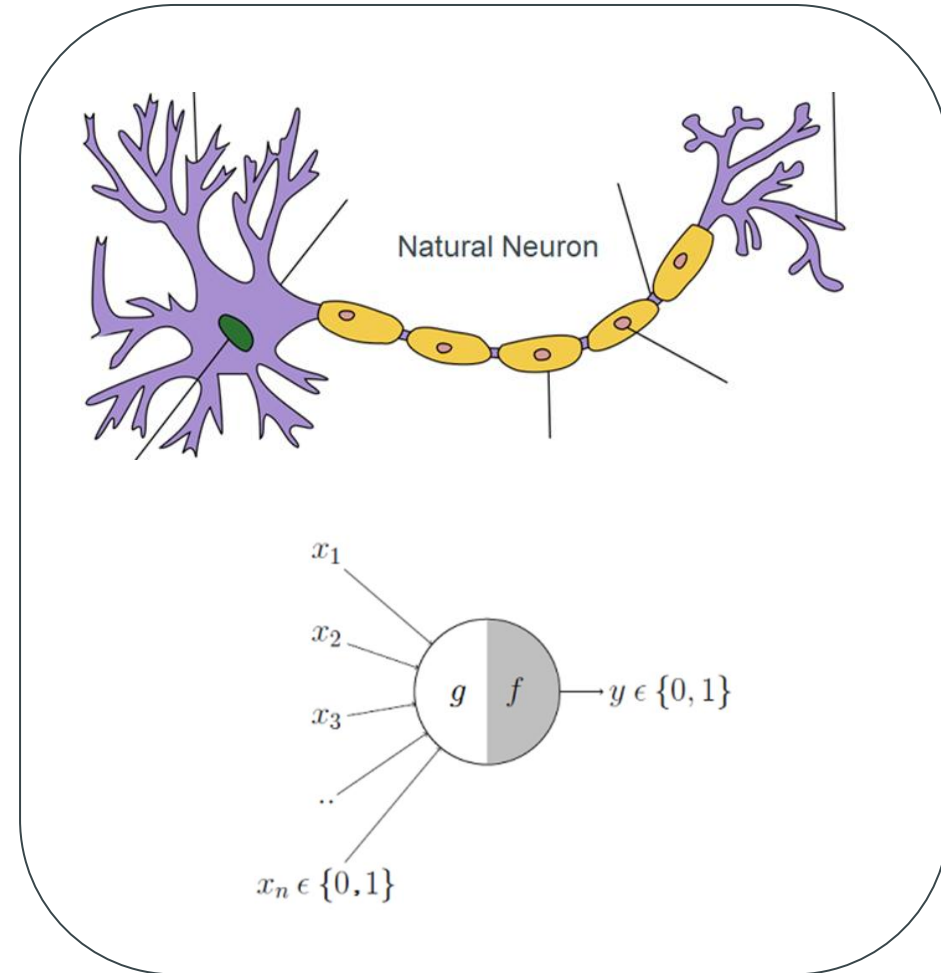
# Neural networks (NNs)

Artificial  
Neural  
Networks

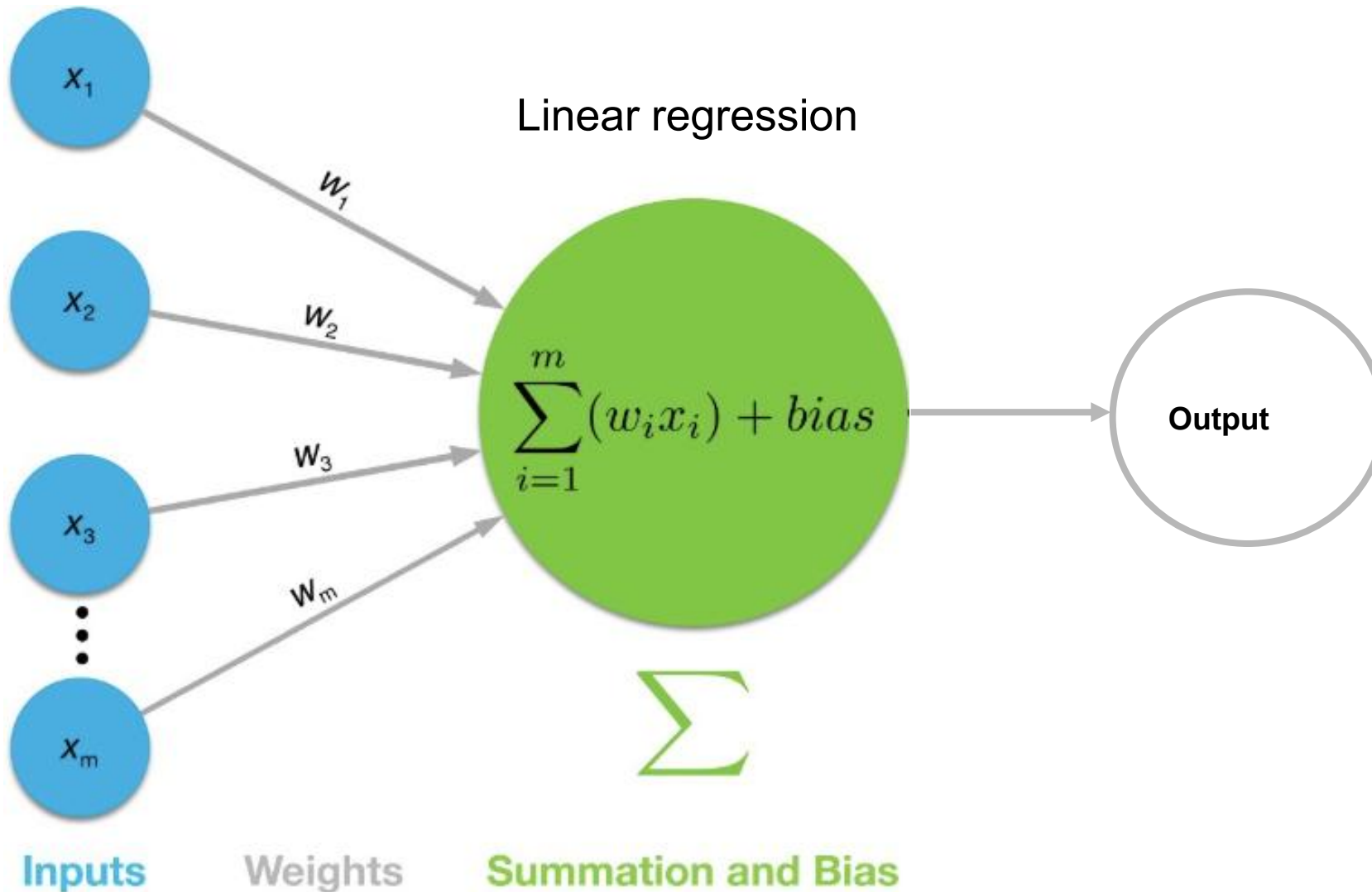
Born in the 1950s  
Died in the 1970s

Reborn in 1980s  
Died again in the 1990s

Reborn in the 2010s  
Currently booming field



# Neural networks (NNs)



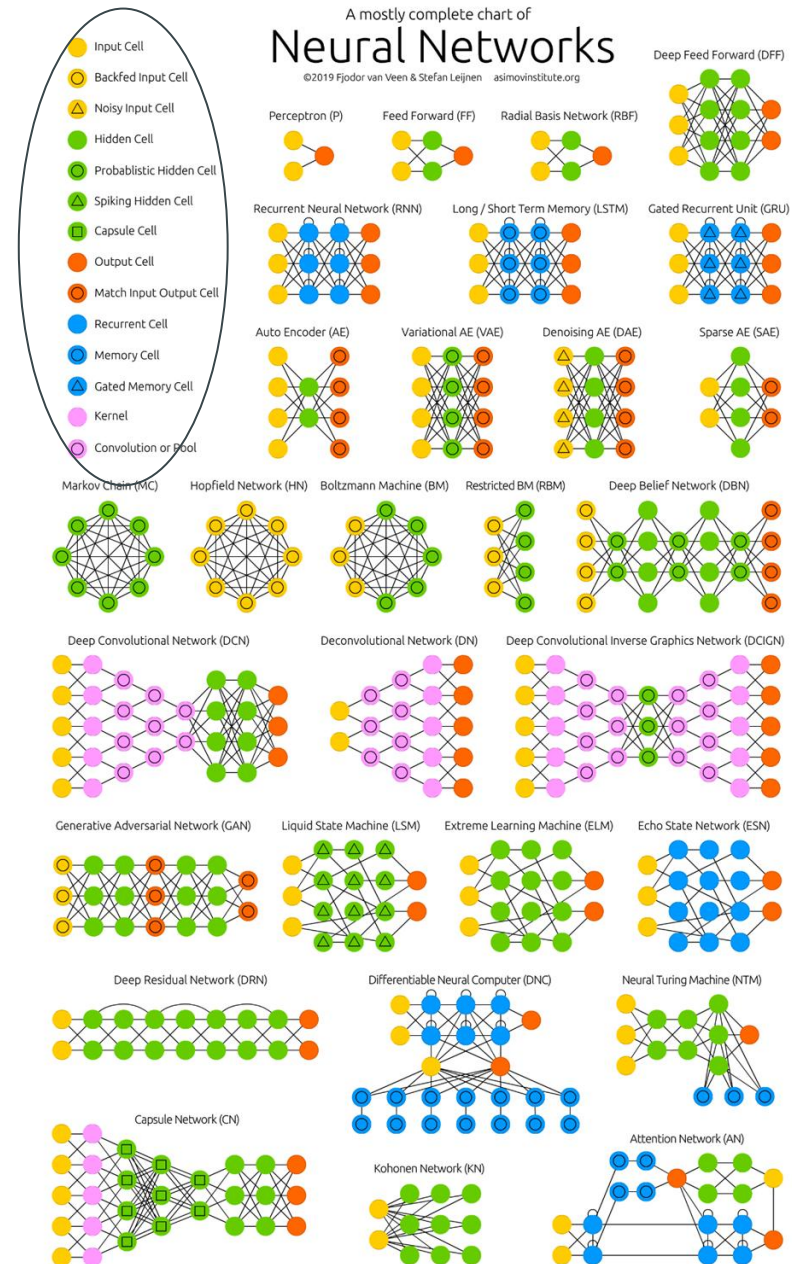
# Deep Neural networks (NNs)

simple  
building  
blocks

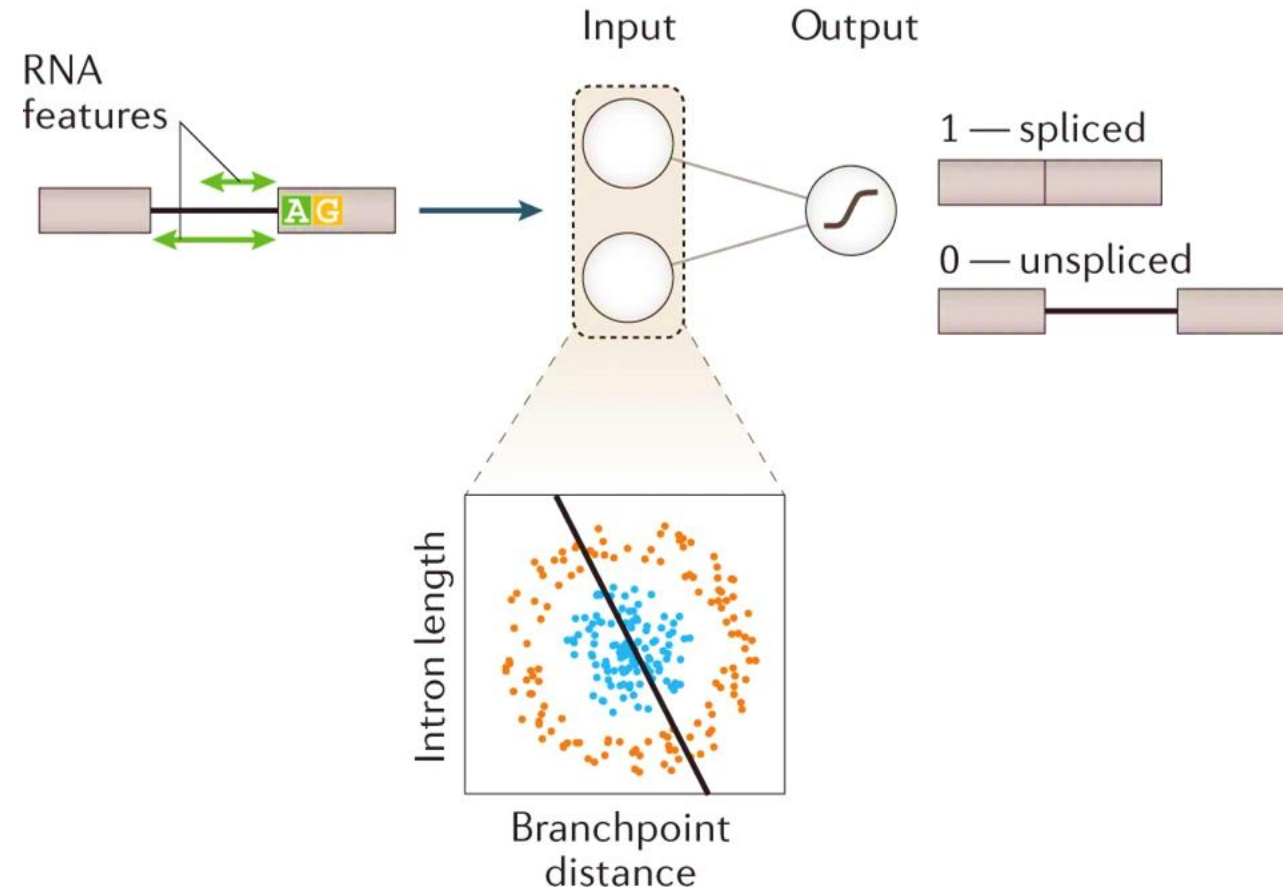
neurons

## Deep Neural Networks

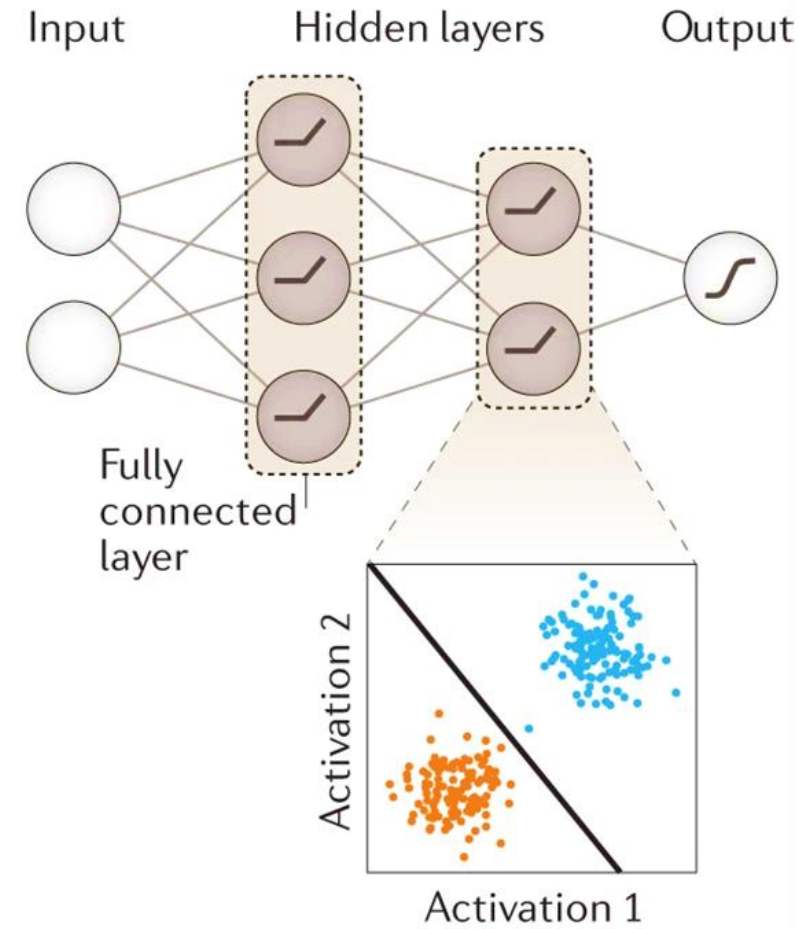
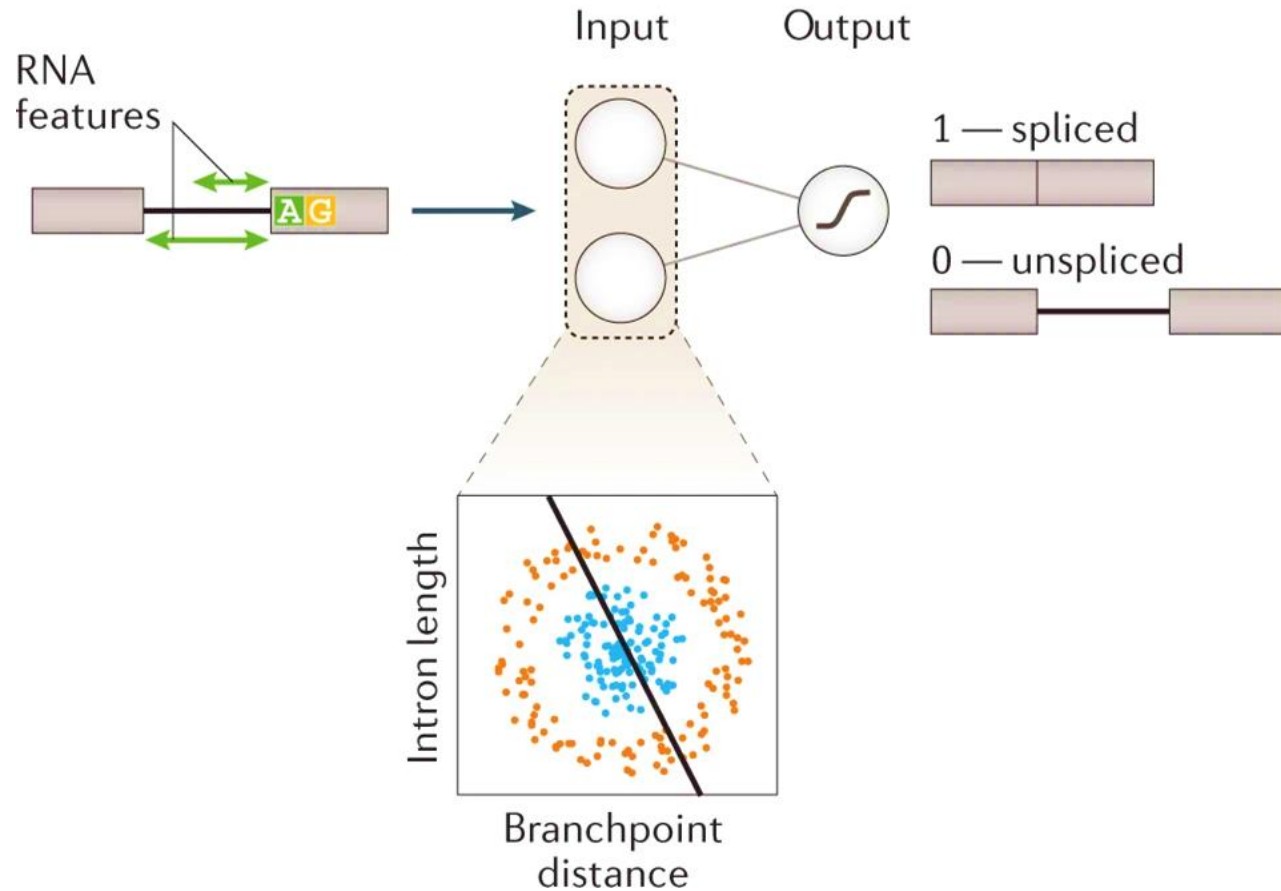
use **several layers** of  
neurons  
to model increasingly  
**complex**  
patterns in data.



# Why neural networks? (NNs)



# Enter Deep Neural Networks (DNNs)





# Convolutional Neural networks (CNNs)

Looks for patterns in input (for e.g., an image), independently of their location

Matrix defines the pattern (filter)

Multiplying the image  $\rightarrow$  number representing the 'fit'



\*

|    |    |    |
|----|----|----|
| -1 | -2 | -1 |
| 0  | 0  | 0  |
| 1  | 2  | 1  |



\*

|    |   |   |
|----|---|---|
| -1 | 0 | 1 |
| -2 | 0 | 2 |
| -1 | 0 | 1 |



|                 |                 |                 |   |   |
|-----------------|-----------------|-----------------|---|---|
| 1 <sub>x1</sub> | 1 <sub>x0</sub> | 1 <sub>x1</sub> | 0 | 0 |
| 0 <sub>x0</sub> | 1 <sub>x1</sub> | 1 <sub>x0</sub> | 1 | 0 |
| 0 <sub>x1</sub> | 0 <sub>x0</sub> | 1 <sub>x1</sub> | 1 | 1 |
| 0               | 0               | 1               | 1 | 0 |
| 0               | 1               | 1               | 0 | 0 |

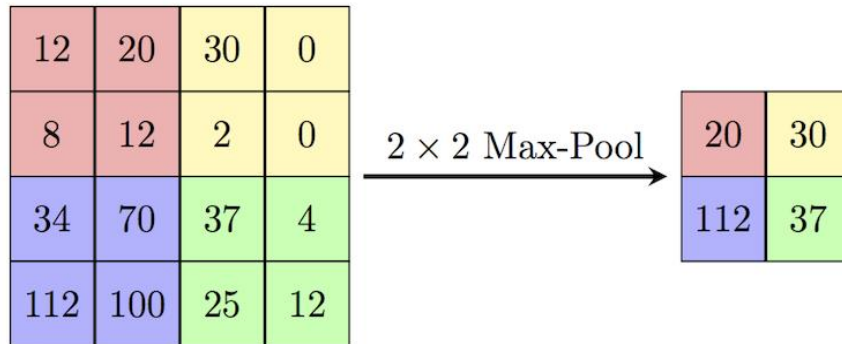
Image

|   |  |  |
|---|--|--|
| 4 |  |  |
|   |  |  |
|   |  |  |

Convolved  
Feature

# Pooling layer

(Max.) - Pooling - taking the most significant areas



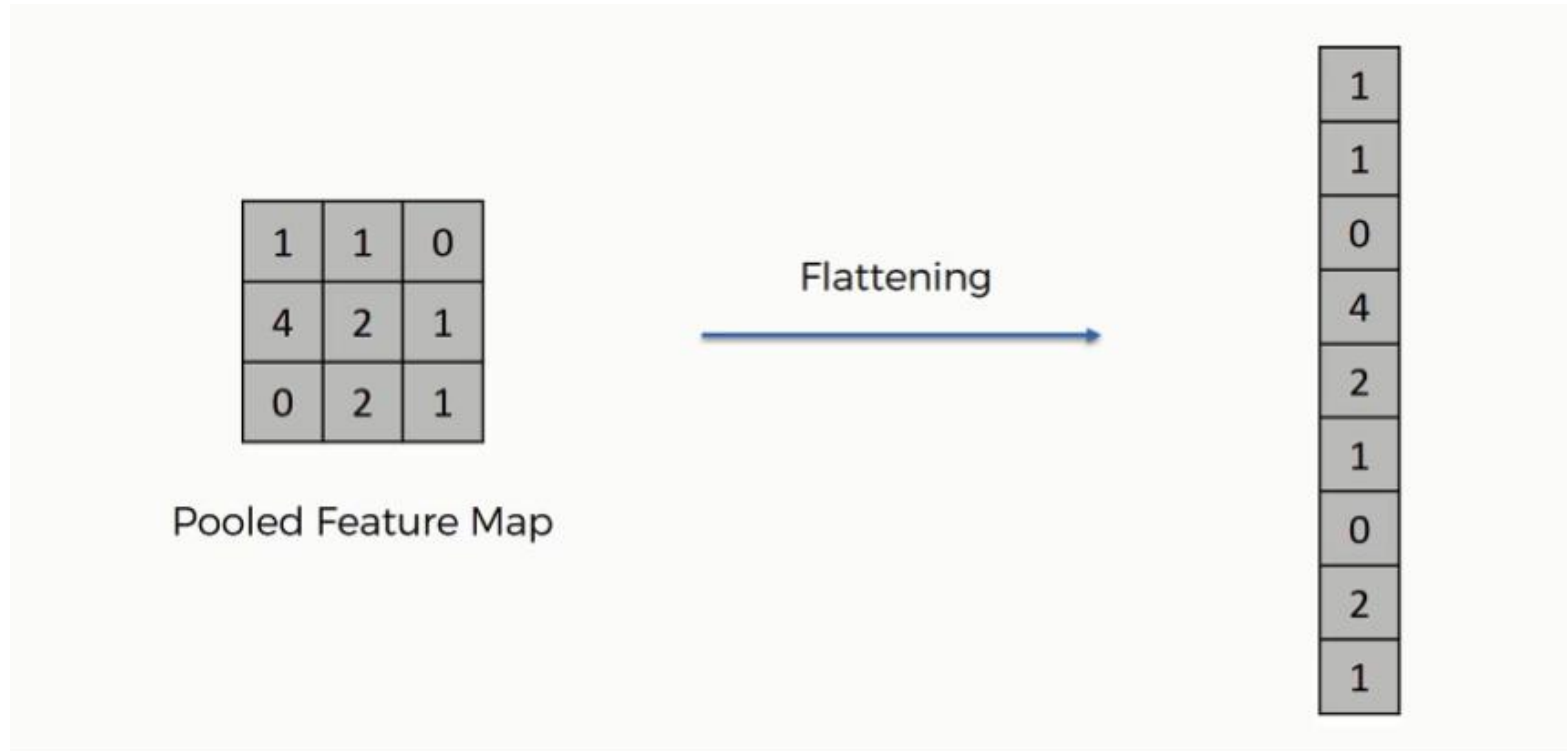
|   |   |   |   |
|---|---|---|---|
| 5 | 5 | 4 | 5 |
| 7 | 6 | 5 | 3 |
| 3 | 5 | 6 | 5 |
| 3 | 4 | 8 | 6 |

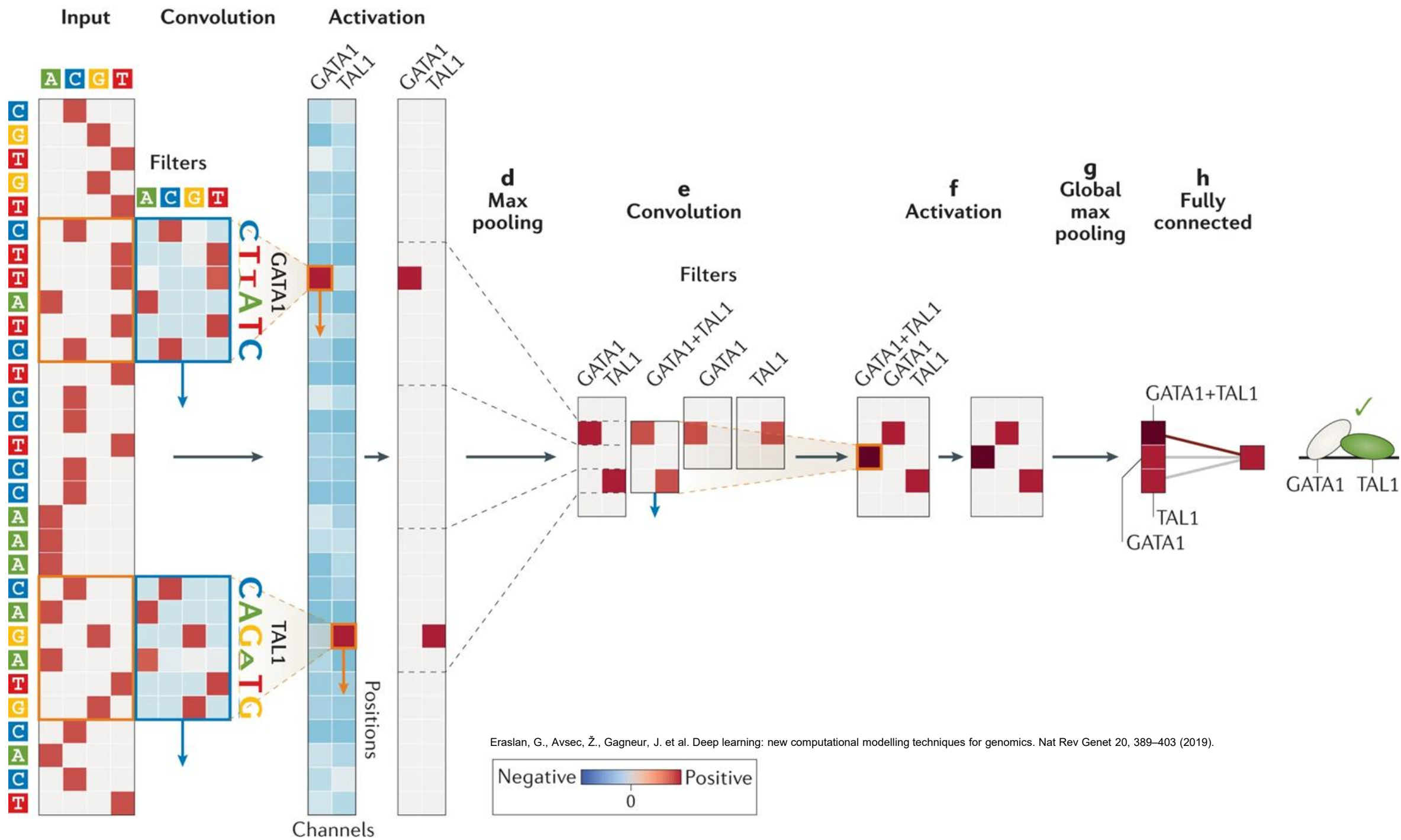
|   |  |
|---|--|
| 7 |  |
|   |  |



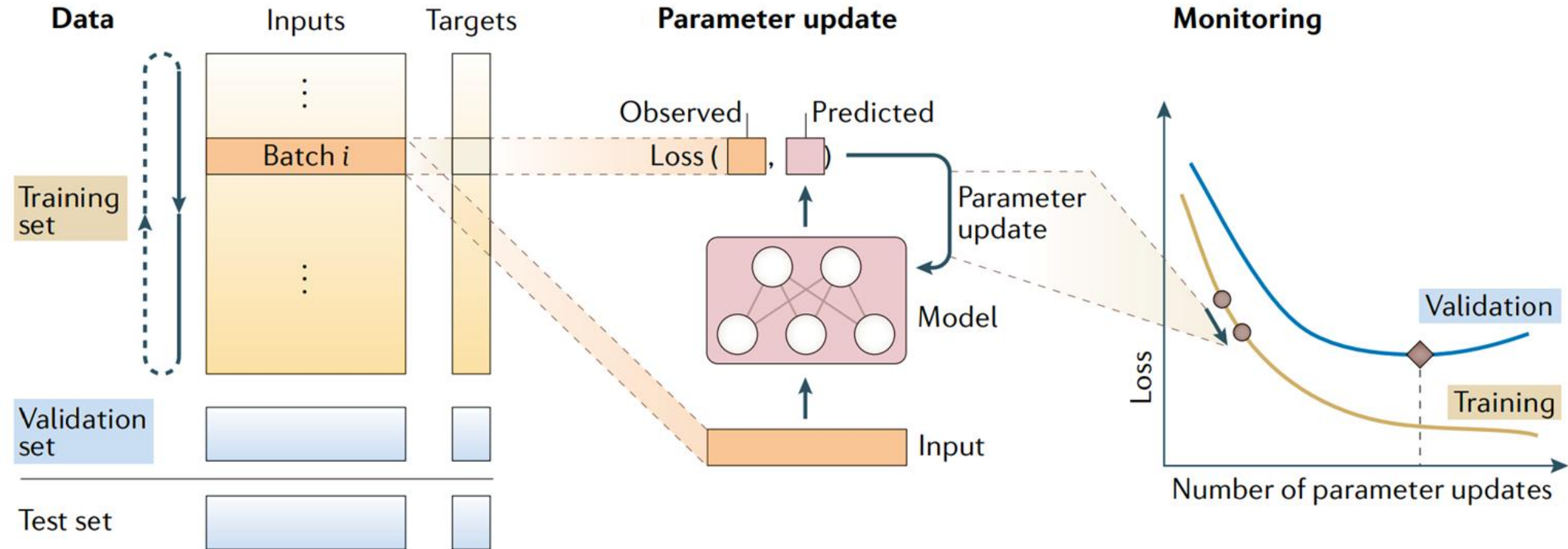
# Flatten layer

Flatten - Spread the pixels and then take combinations of these

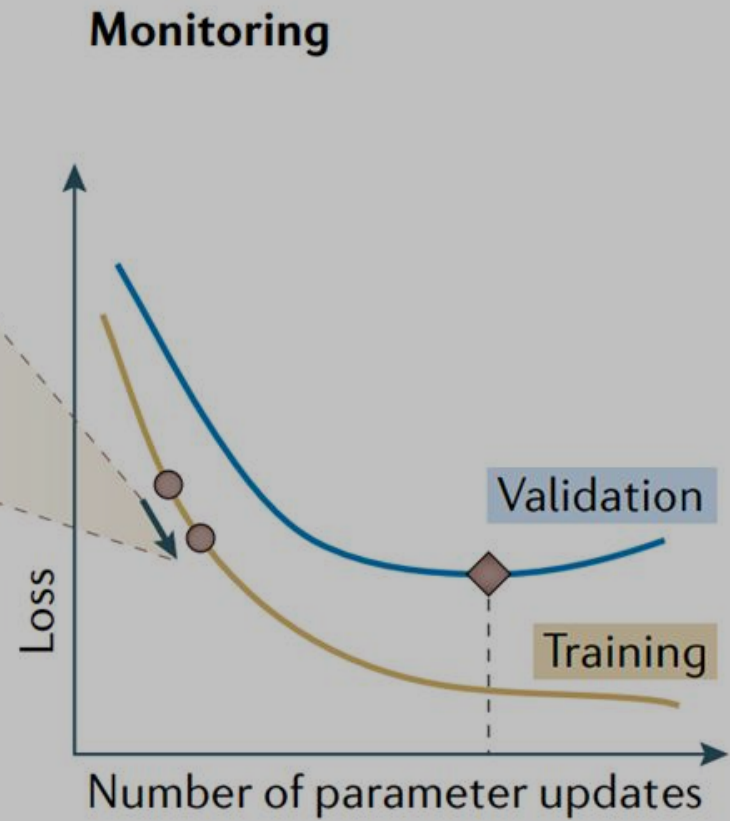
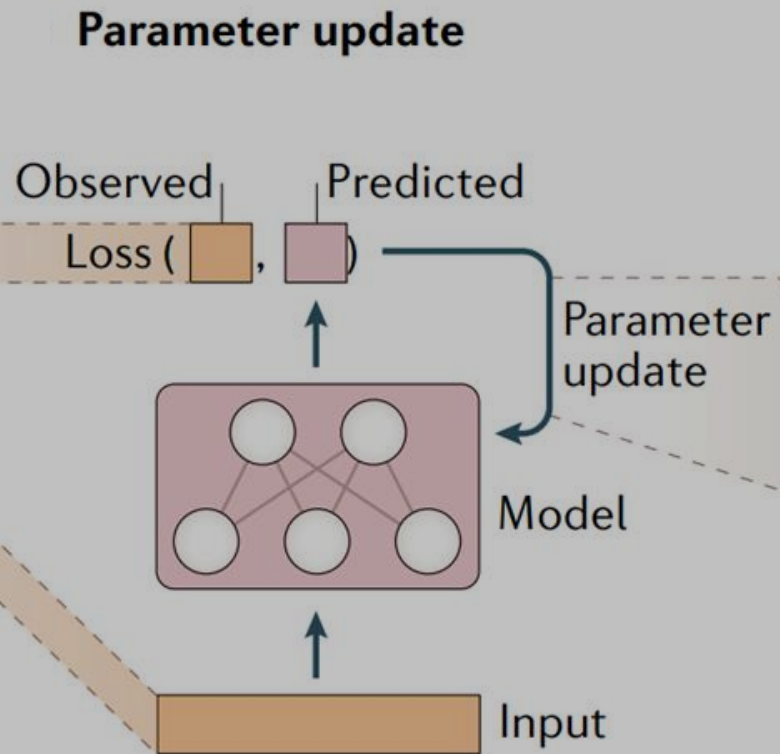
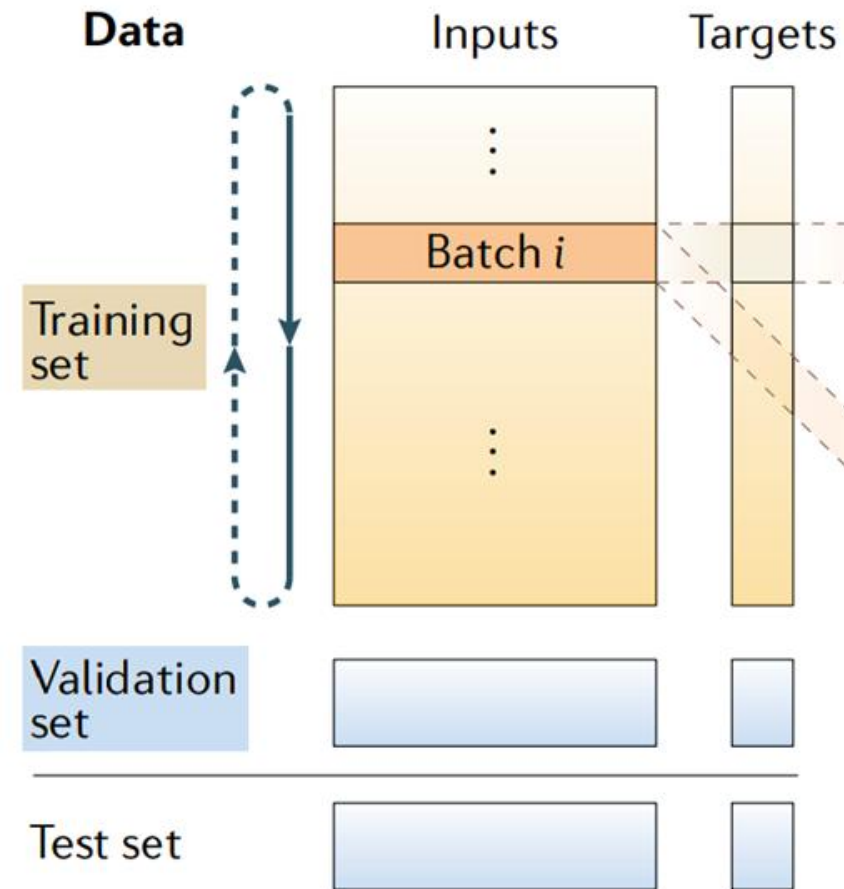


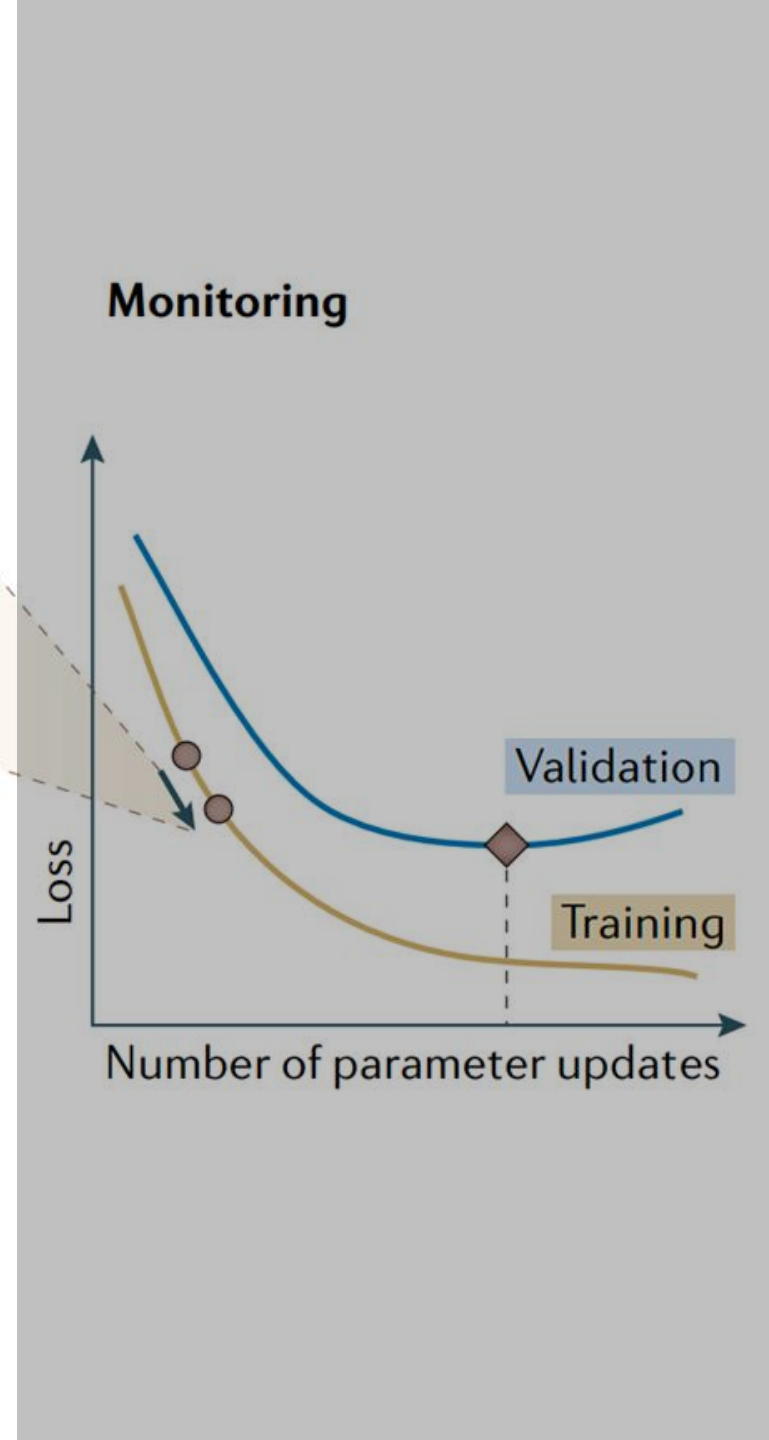
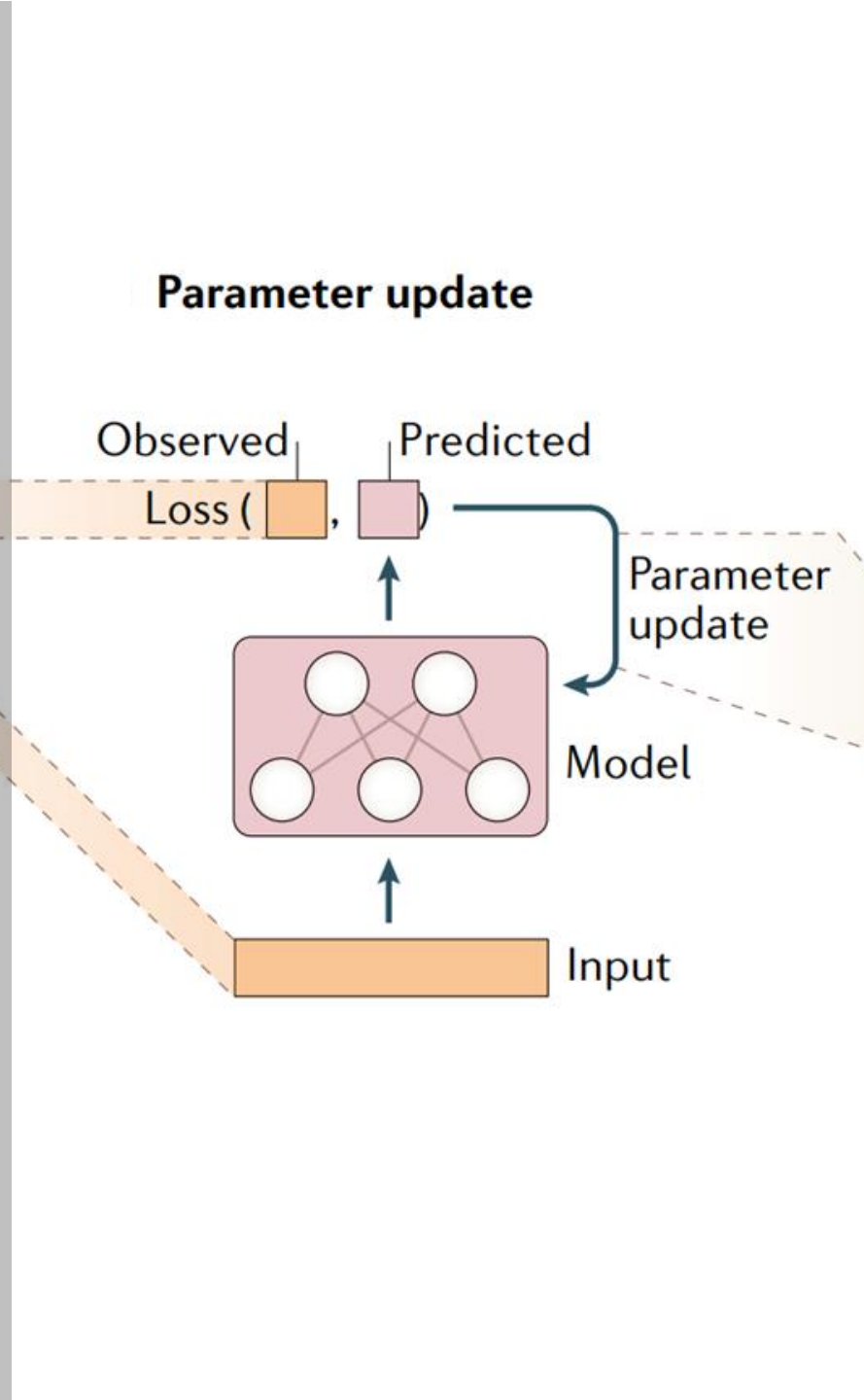
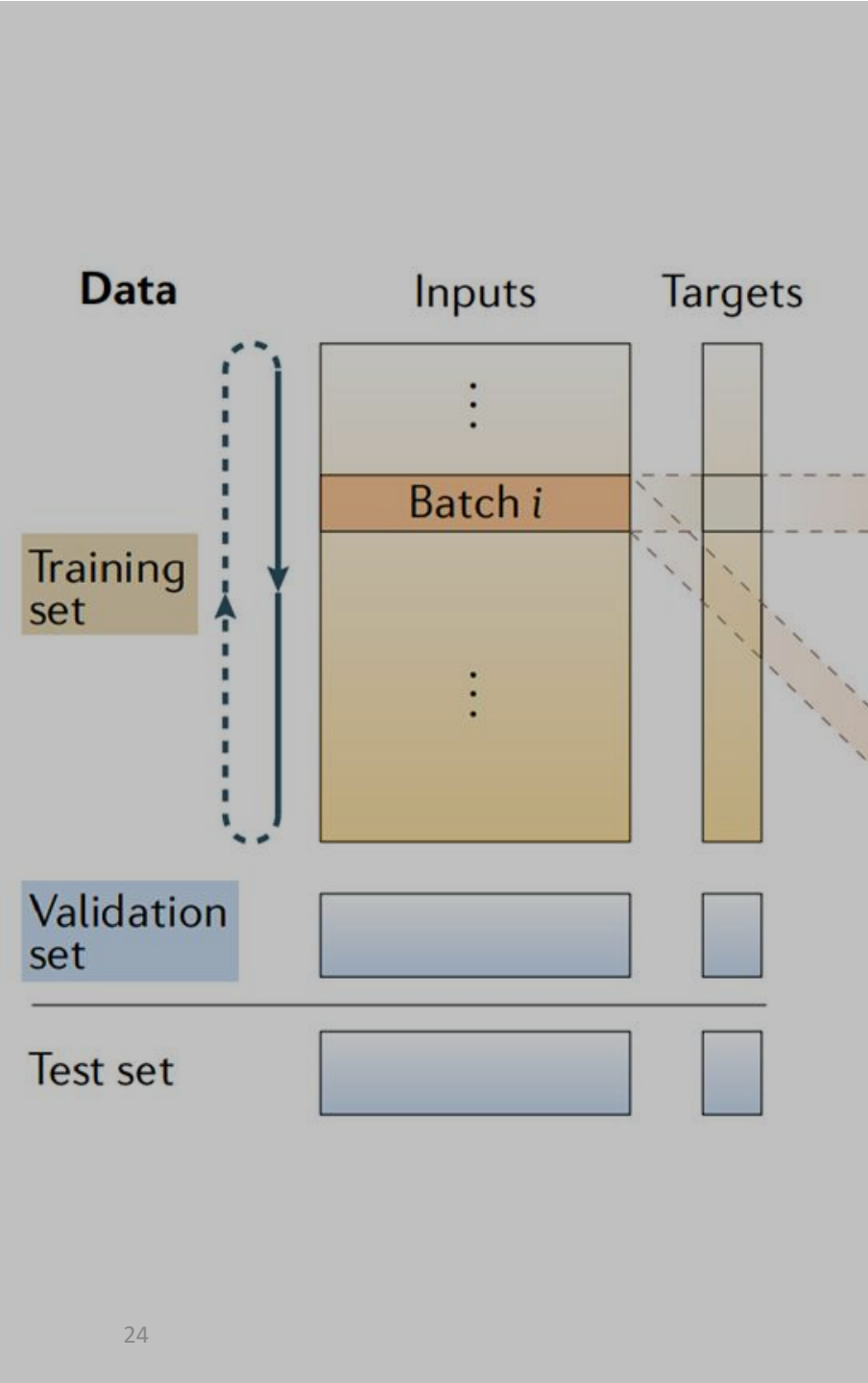


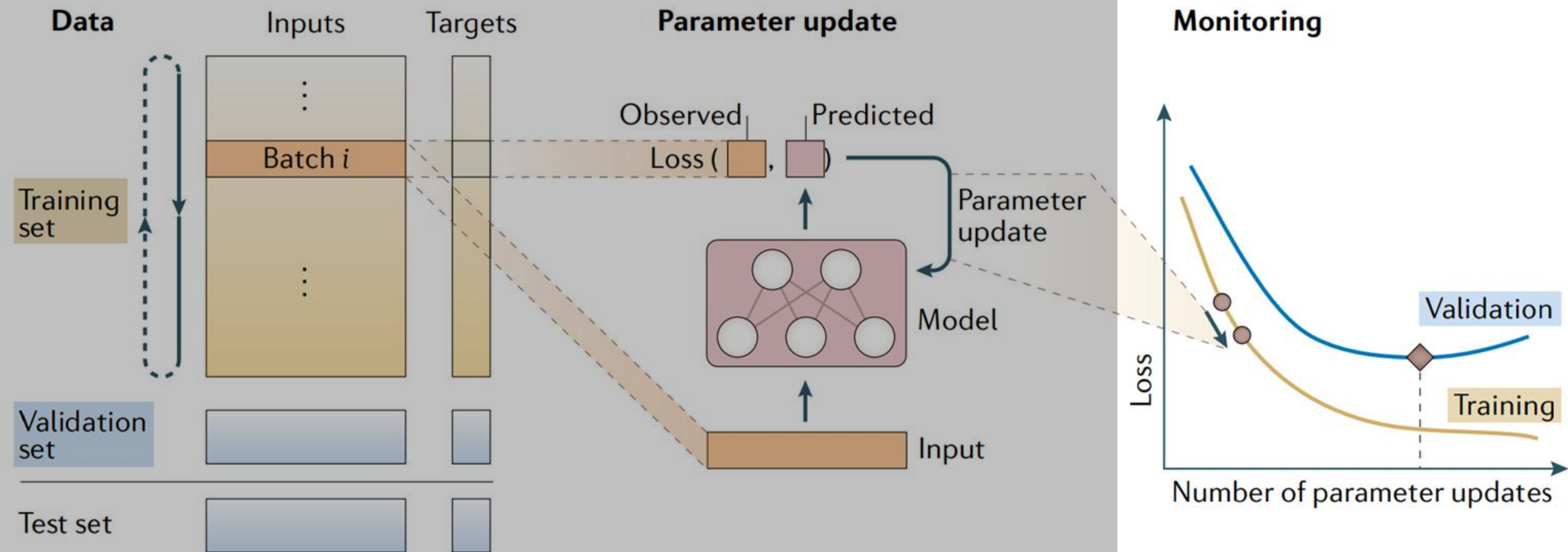
# Model training



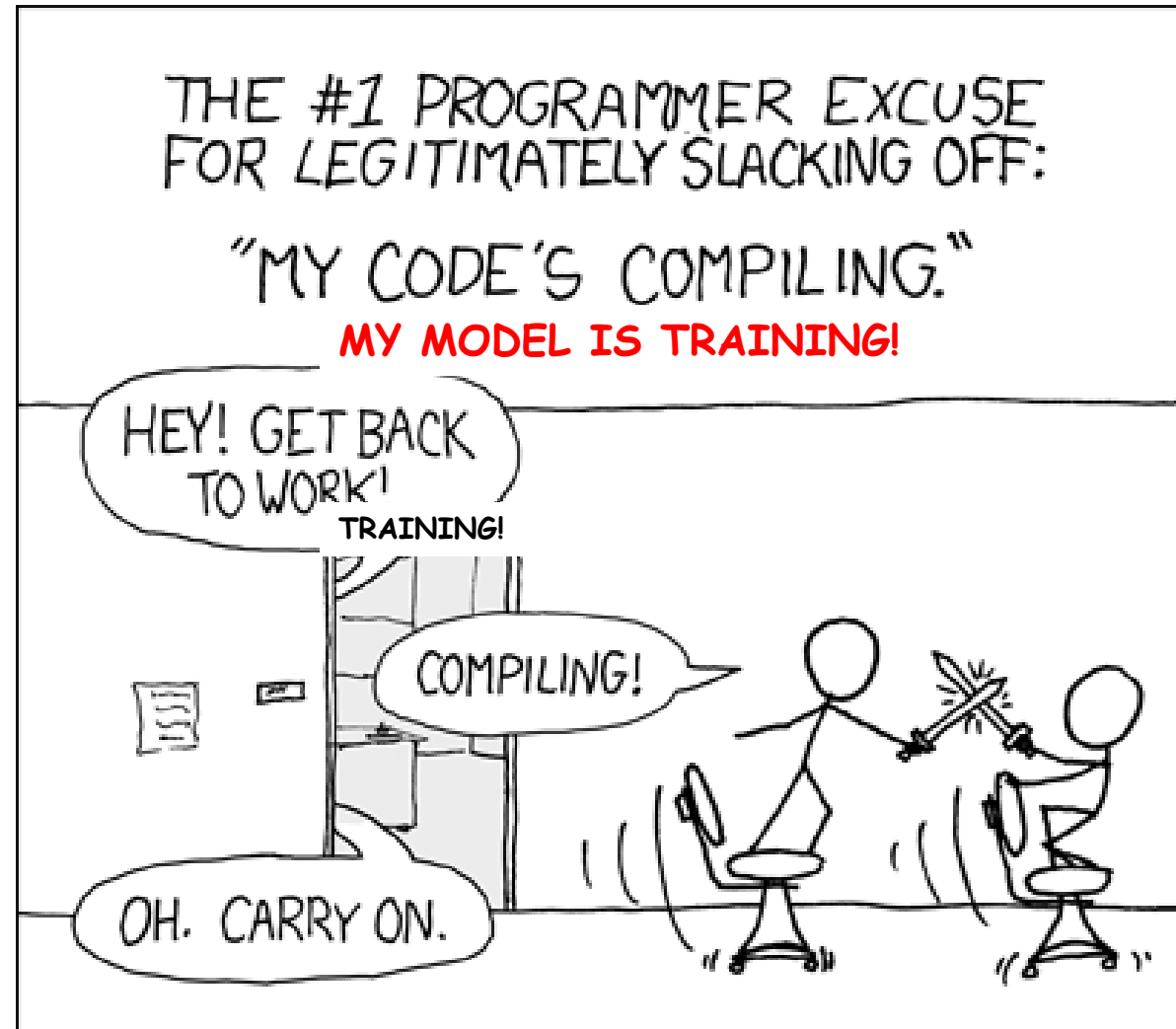
# Model training







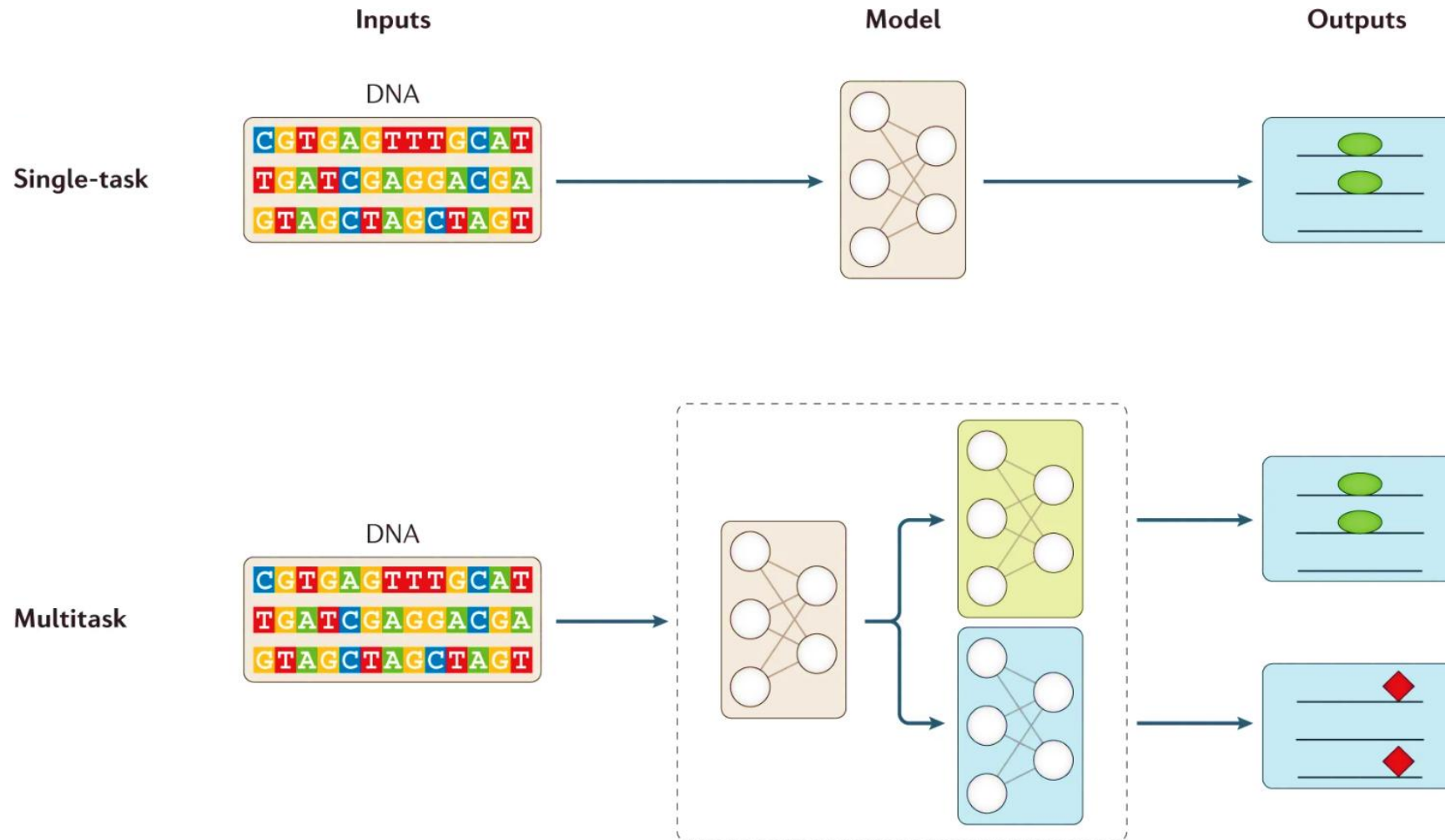
Hmm...



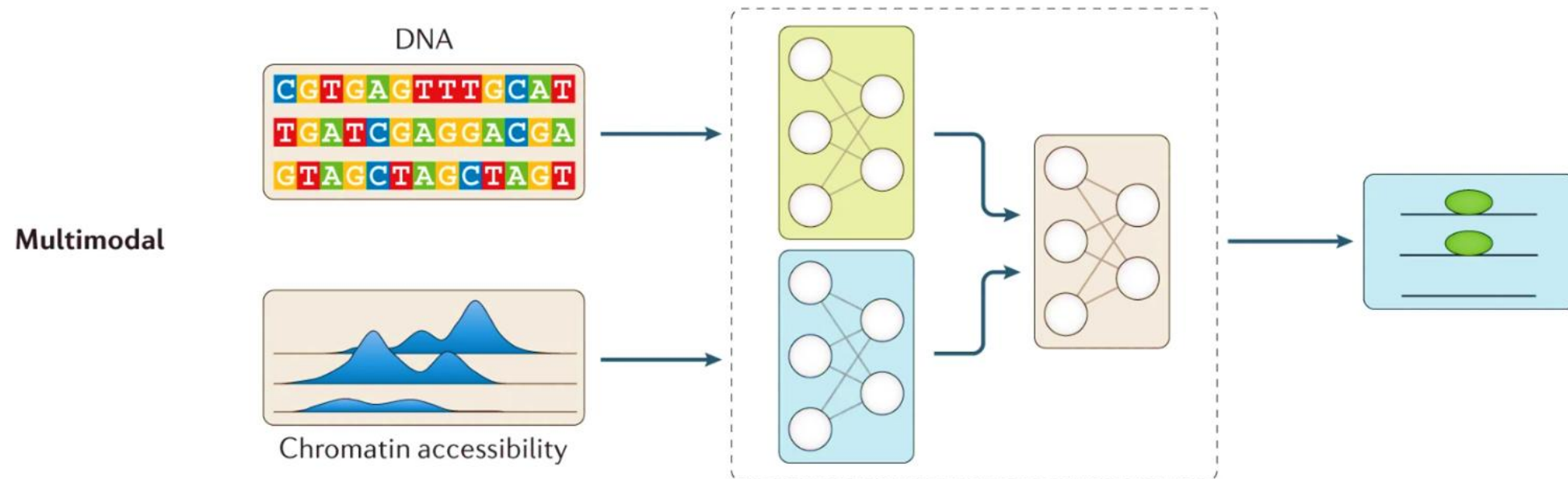
<https://xkcd.com/303/>



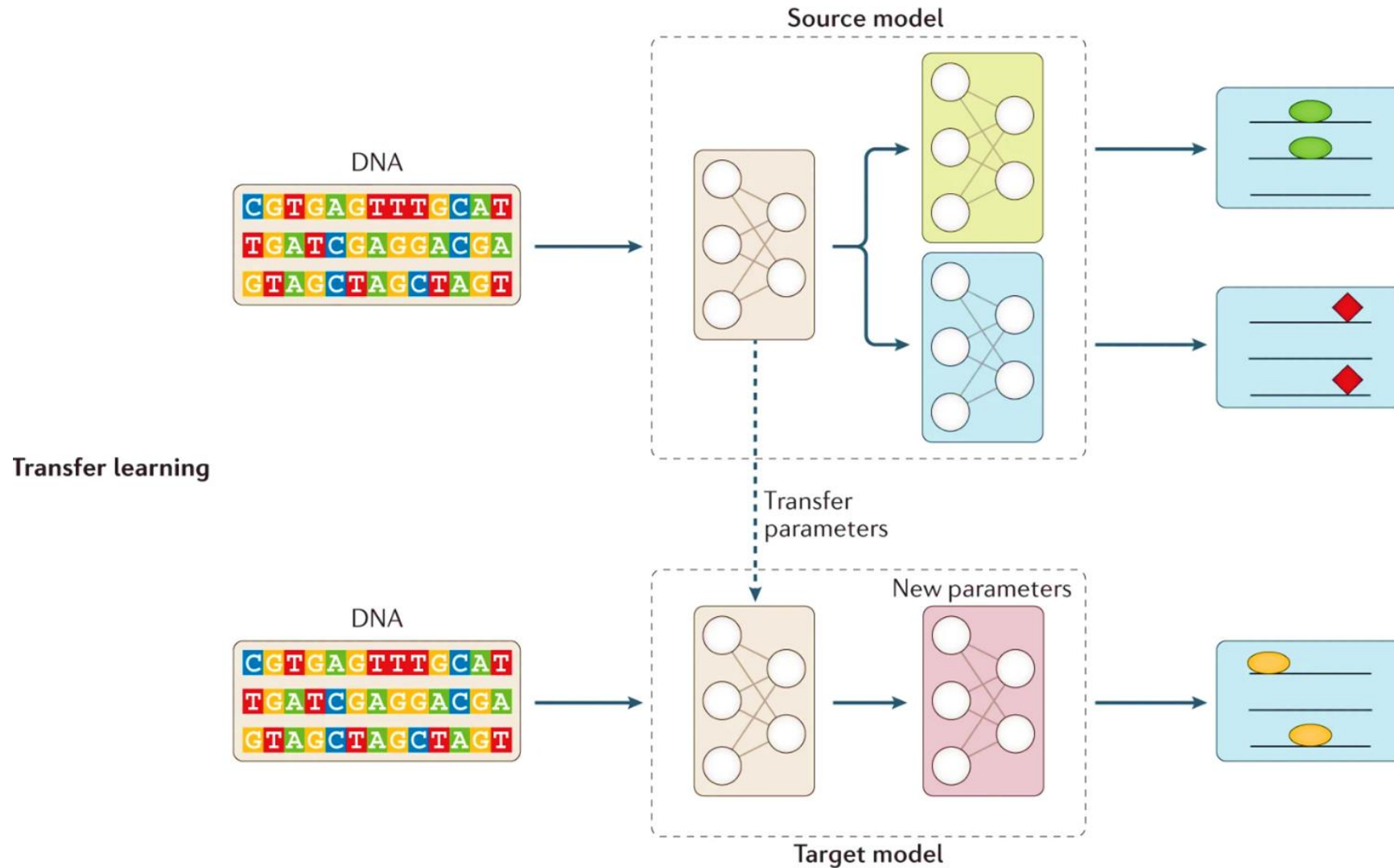
# Model training – How to *stir the pile*?



# Model training – How to *stir the pile*?



# Model training – How to *stir the pile*?



## Useful resources

Goodfellow, I., Bengio, Y. & Courville, A. *Deep Learning* (MIT Press, 2016). This textbook covers theoretical and practical aspects of deep learning with introductory sections on linear algebra and machine learning.

Eraslan, G., Avsec, Ž., Gagneur, J. *et al.* Deep learning: new computational modelling techniques for genomics. *Nat Rev Genet* 20, 389–403 (2019).

<https://huggingface.co/>

The AI community building the future.

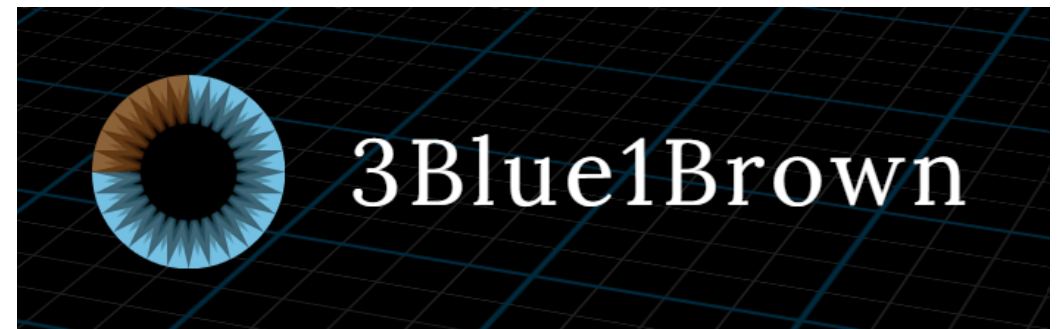


**Kipoi**

<https://kipoi.org/>

Model Zoo for genomics

<https://www.3blue1brown.com/>



## Useful resources (2)

<https://www.kaggle.com/>



### Critical Assessment of Genome Interpretation

Challenges

Meetings

Publications

Organization

Contact Us

<https://genomeinterpretation.org/>

### Challenges

<https://paperswithcode.com/>



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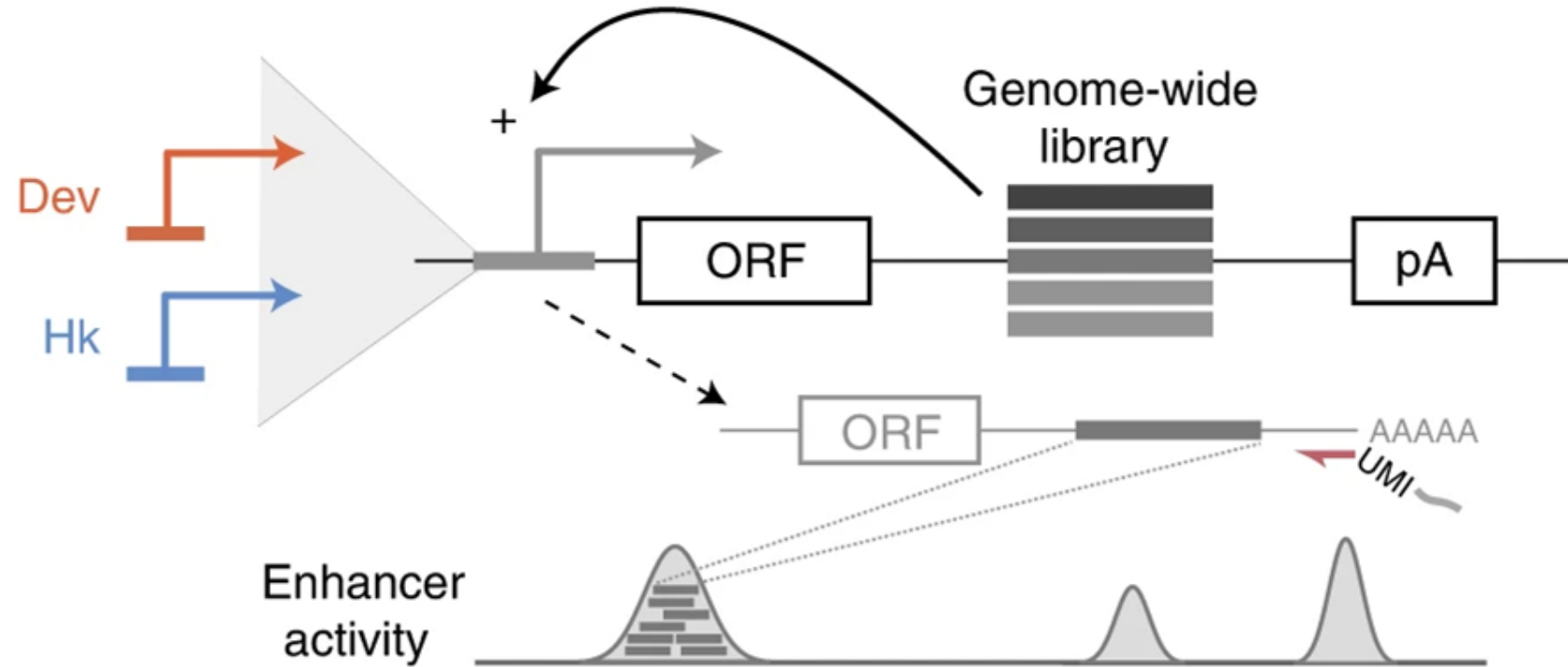
Donate ▾

[https://sagebionetworks.org/tools\\_resources/challenges/](https://sagebionetworks.org/tools_resources/challenges/)

Challenge Platform

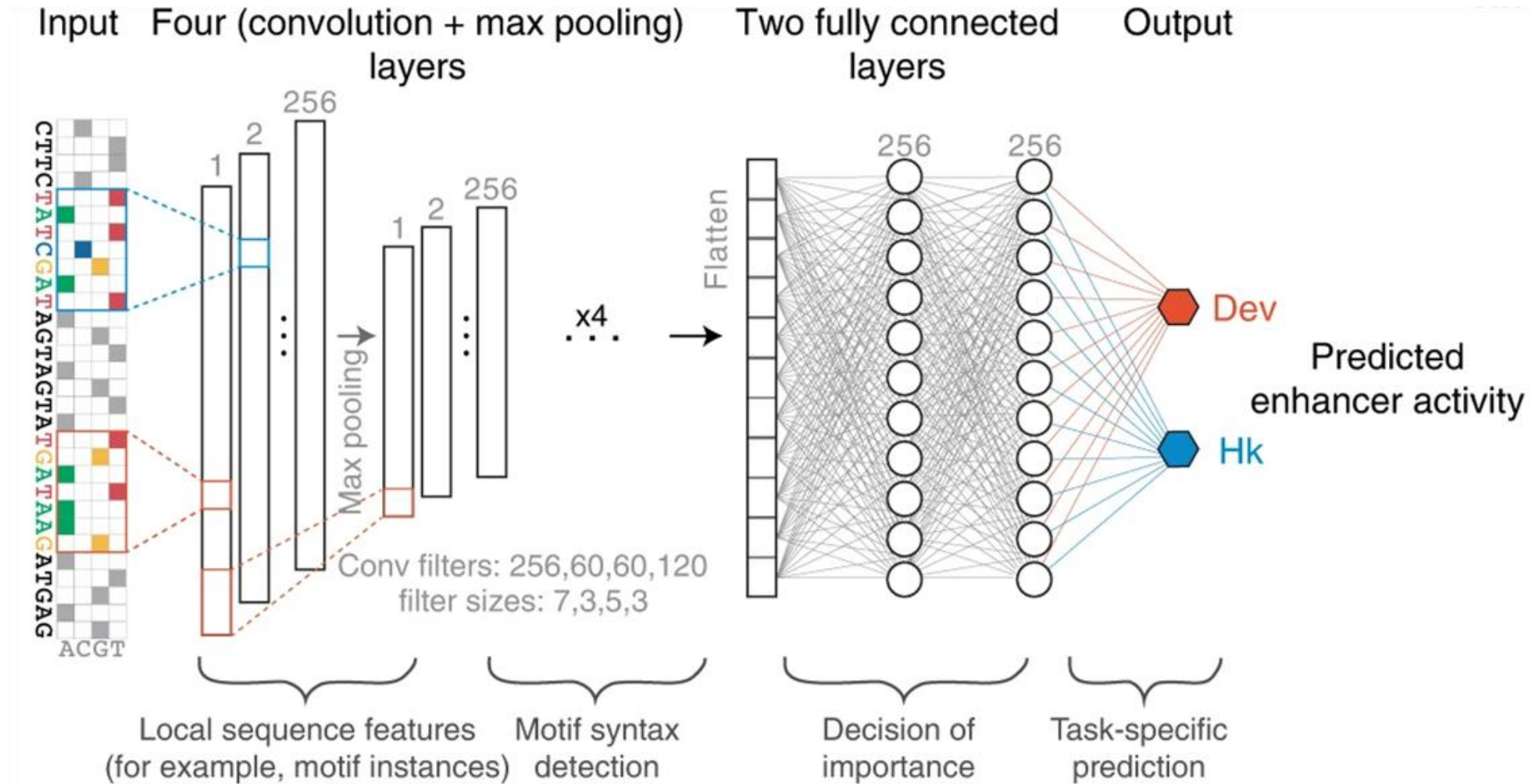


# Example: DeepSTARR



de Almeida, B.P., Reiter, F., Pagani, M. *et al.* DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* 54, 613–624 (2022).

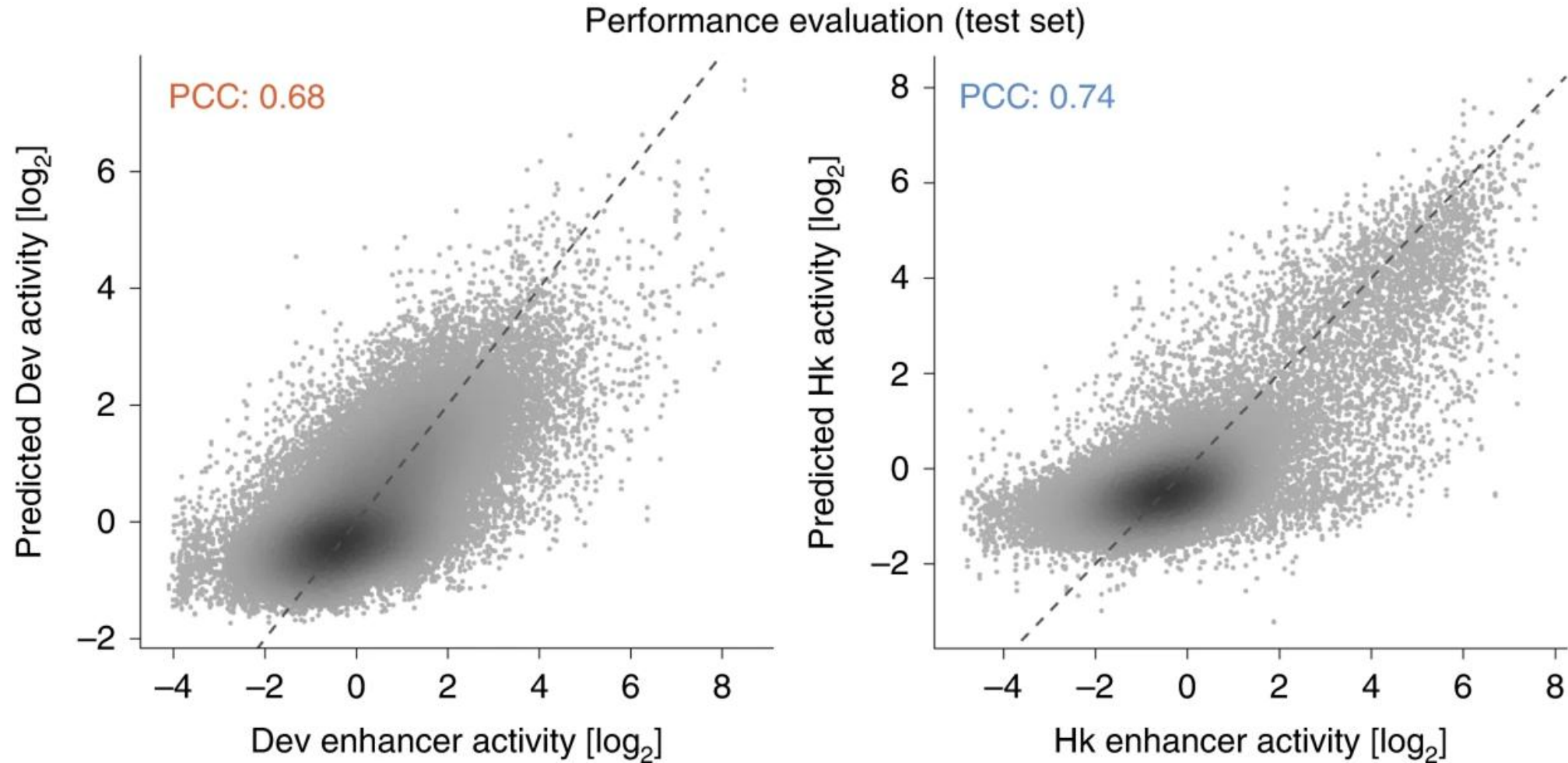
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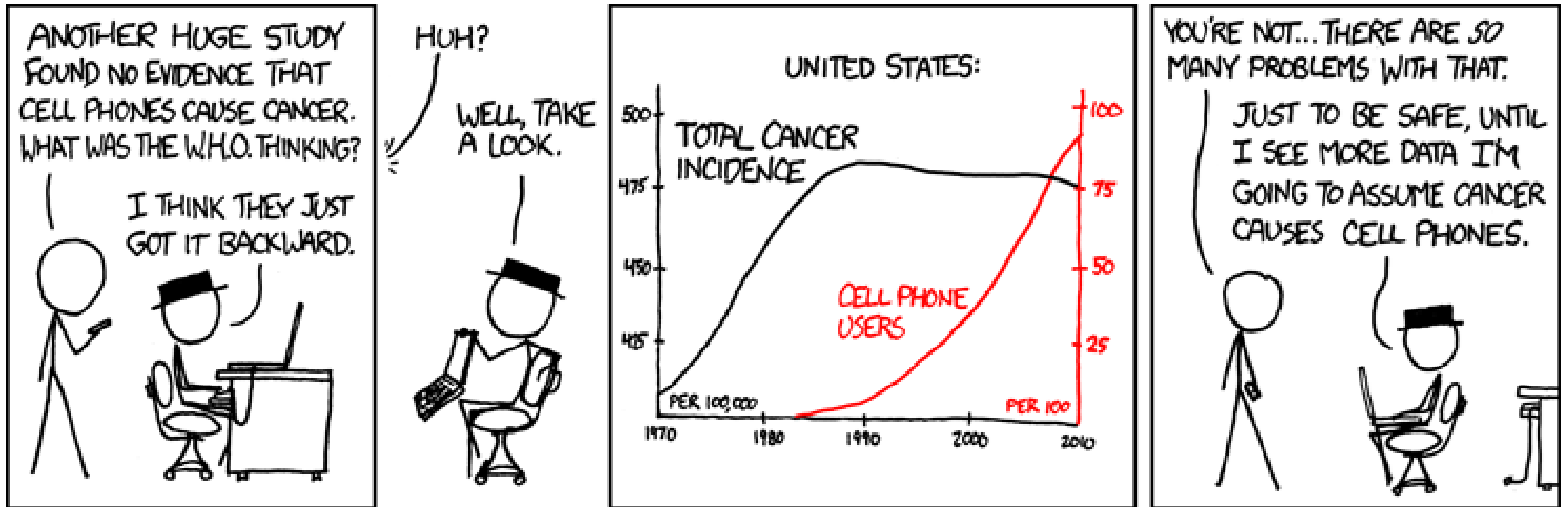


# Example: DeepSTARR



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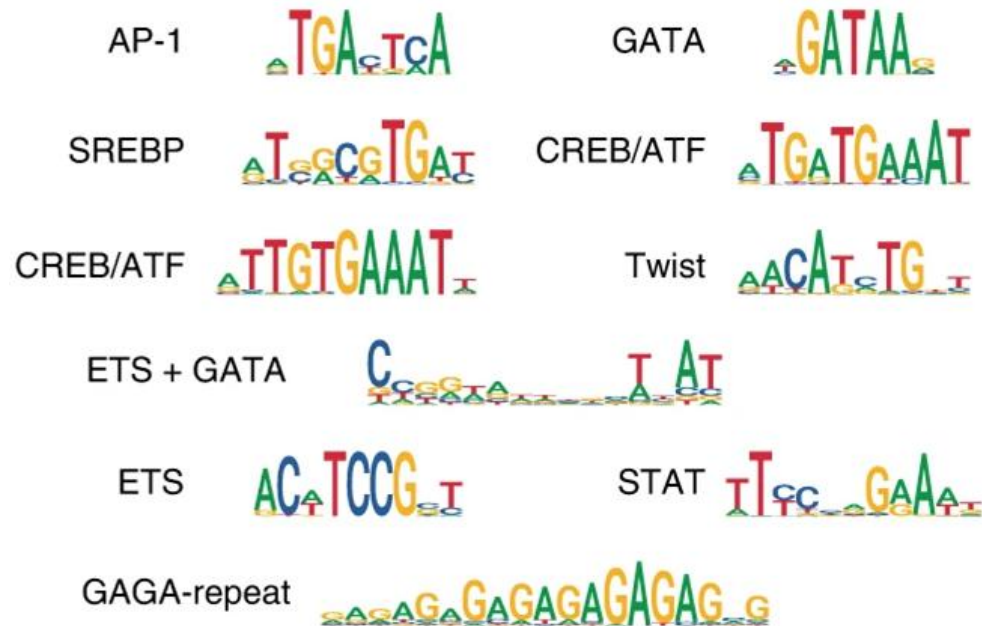
# Model interpretation



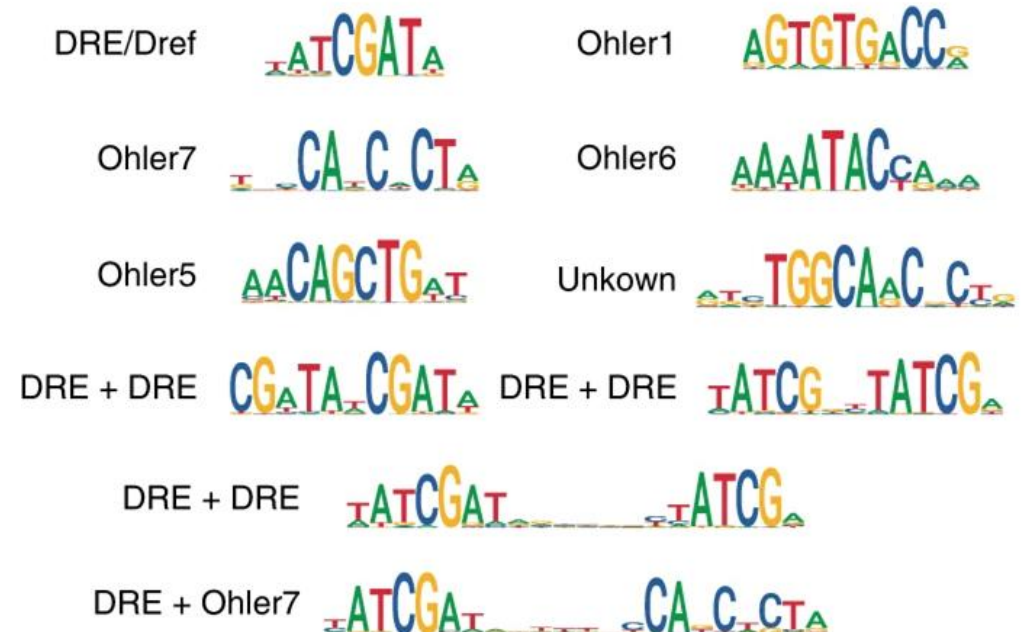
<https://xkcd.com/925/>

# Example: DeepSTARR

Dev TF motifs (DeepSTARR)

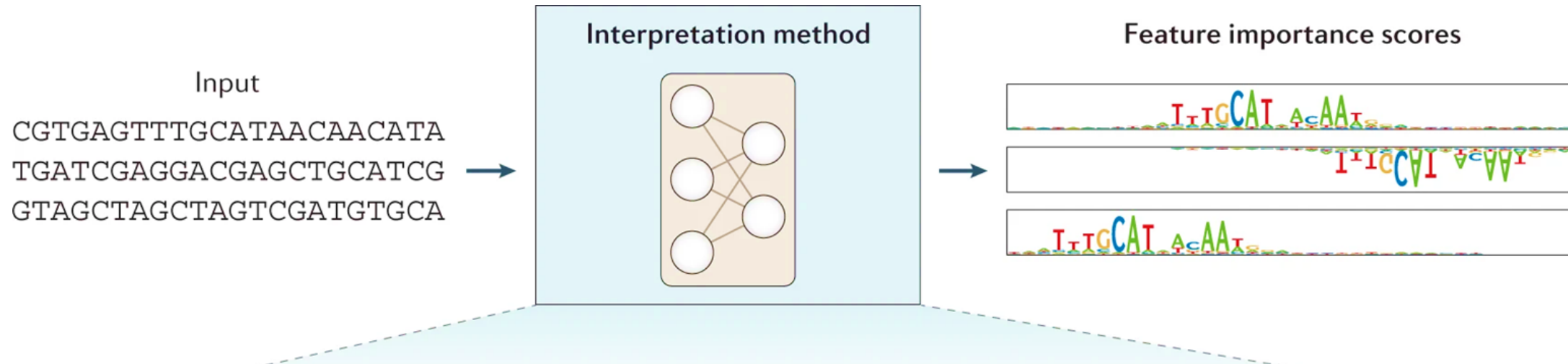


Hk TF motifs (DeepSTARR)



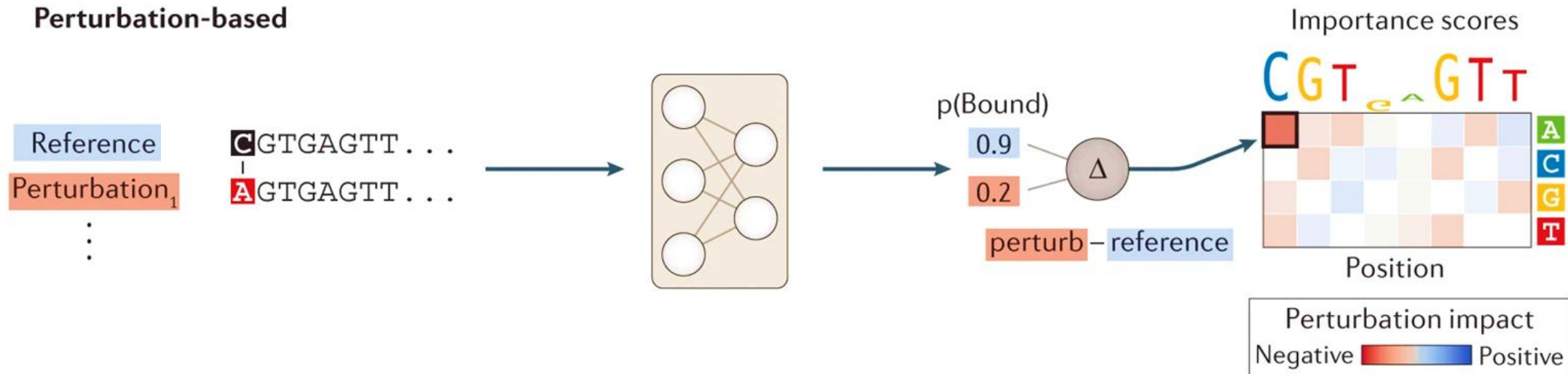
de Almeida, B.P., Reiter, F., Pagani, M. *et al.* DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* 54, 613–624 (2022).

# Model Interpretation - the idea



# Model Interpretation - ISM

## Perturbation-based



# Model Interpretation - ISM

## Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks

David R. Kelley<sup>1</sup>, Jasper Snoek<sup>2</sup> and John L. Rinn<sup>1</sup>

+ Author Affiliations

Corresponding author: [dkelley@fas.harvard.edu](mailto:dkelley@fas.harvard.edu)





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[kircherlab.github.io](https://kircherlab.github.io)



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# Hands-On sessions: Model training



**ISMB-ECCB 2025: Tutorial IP2 (MPRAs and iGVF Resource)**

Arjun Devadas, Max Schubach  
Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH  
July 20, 2025



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Schleswig-Holstein

**BIH** Berlin Institute  
of Health  
@Charité

| <u>P-VALUE</u> | <u>INTERPRETATION</u>                                  |
|----------------|--------------------------------------------------------|
| 0.001          | HIGHLY SIGNIFICANT                                     |
| 0.01           |                                                        |
| 0.02           |                                                        |
| 0.03           |                                                        |
| 0.04           | SIGNIFICANT                                            |
| 0.049          |                                                        |
| 0.050          | OH CRAP. REDO CALCULATIONS.                            |
| 0.051          | ON THE EDGE OF SIGNIFICANCE                            |
| 0.06           |                                                        |
| 0.07           | HIGHLY SUGGESTIVE, SIGNIFICANT AT THE $P < 0.10$ LEVEL |
| 0.08           |                                                        |
| 0.09           |                                                        |
| 0.099          | HEY, LOOK AT THIS INTERESTING SUBGROUP ANALYSIS        |
| $\geq 0.1$     |                                                        |

## Hands-on: Training a sequence-based model



[https://github.com/kircherlab/ISMB-2025\\_IGVF-MPRA-Tutorial](https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial)

**Go to:** 05\_sequence\_models/01\_training\_sequence\_model.ipynb

**Klick:** Open in Colab

# Discussion of pre-training and fine-tuning approaches



*ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)*

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Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

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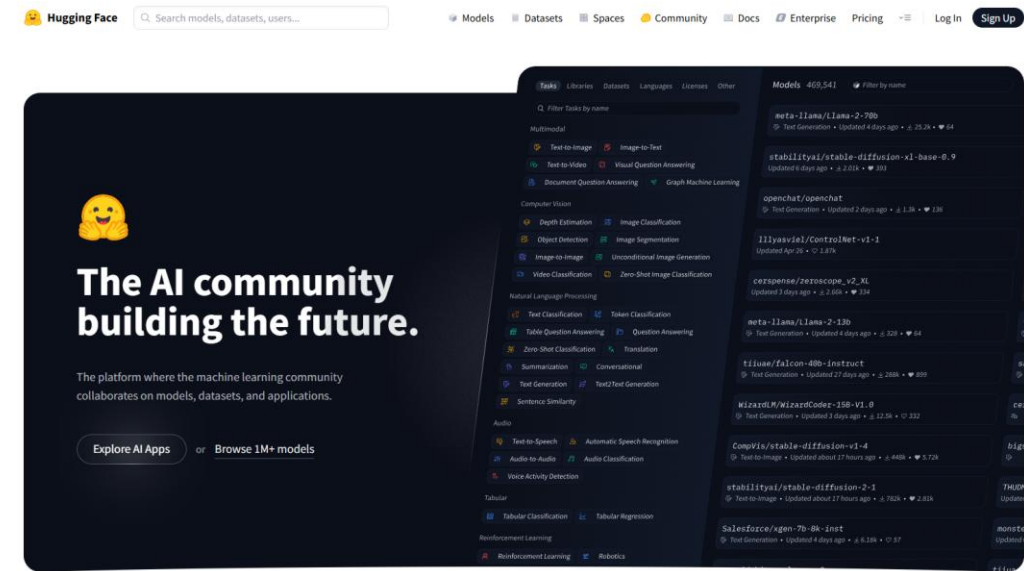


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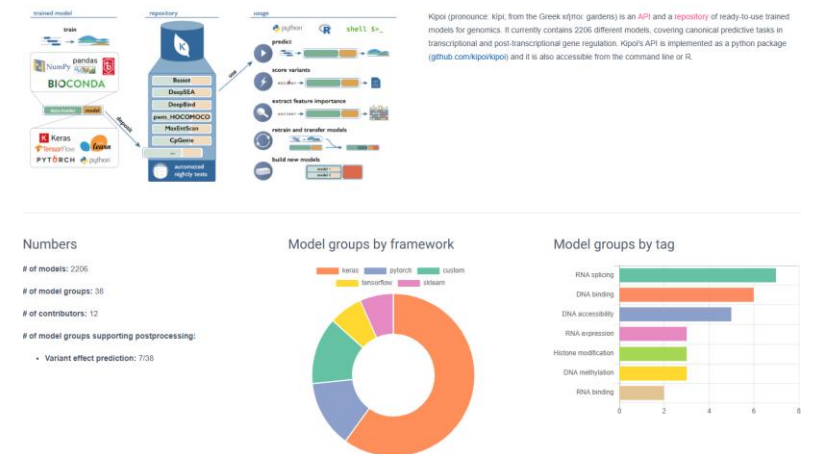
# Benefiting from available resources

- **Accelerated Development:** Leverage pre-trained models to jump-start projects, reducing prototyping and training time.
- **Resource Efficiency, Flexibility and Adaptability:** Utilize models that have already been trained on massive datasets, saving on computational costs and data collection efforts.
- **Robust Feature Extraction:** Benefit from learned representations that capture complex patterns, making fine-tuning more effective for specific tasks.
- **Standardized Benchmarks:** Use well-established models as baselines to ensure reproducibility and make comparative evaluations more straightforward.



<https://huggingface.co/>

Kipoi: Model zoo for genomics

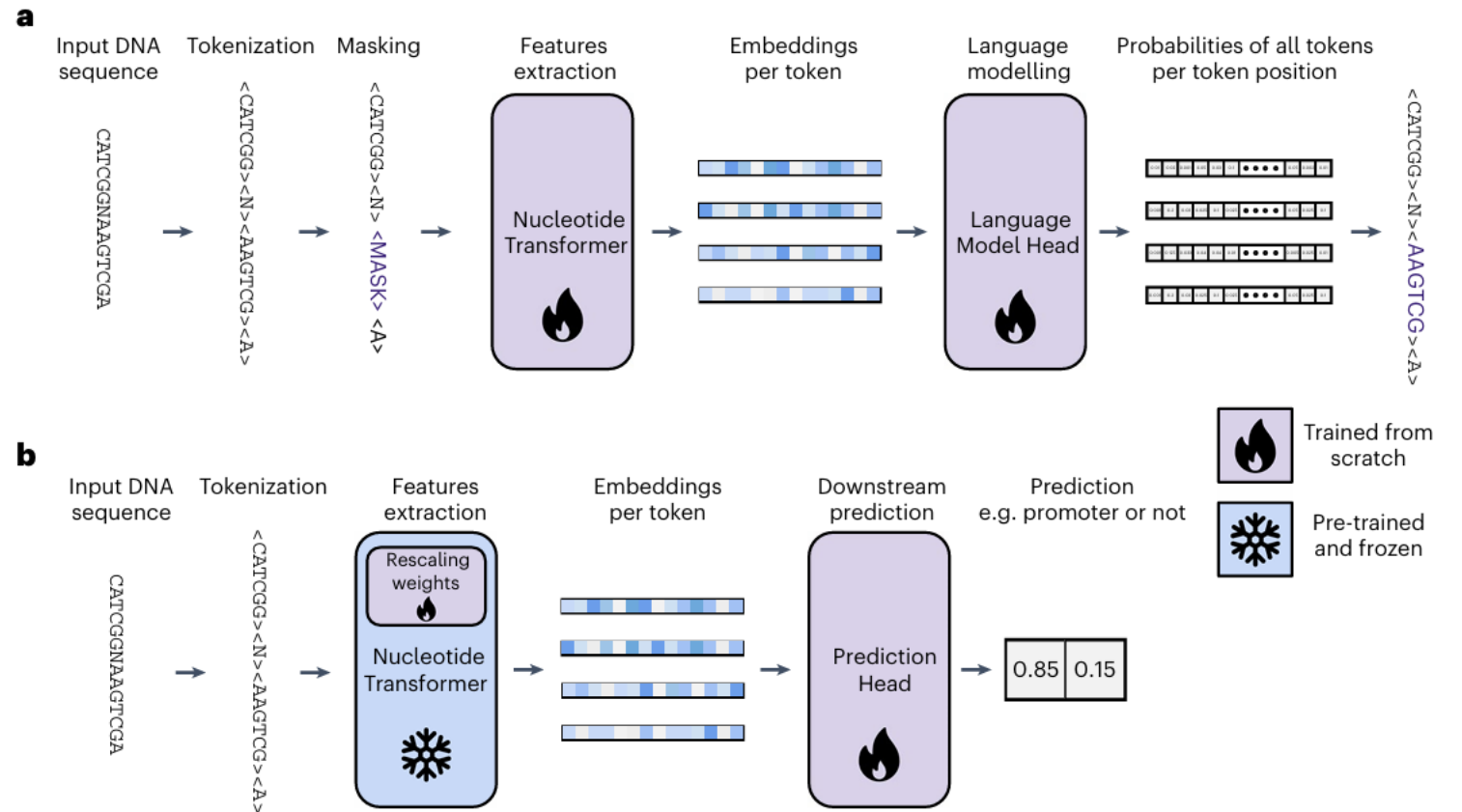


<https://kipoi.org/>

# Foundation models like DNABERT, HyenaDNA, NT or Enformer

Foundation models are pre-trained on massive datasets, converting raw nucleotide sequences into rich, context-aware representations

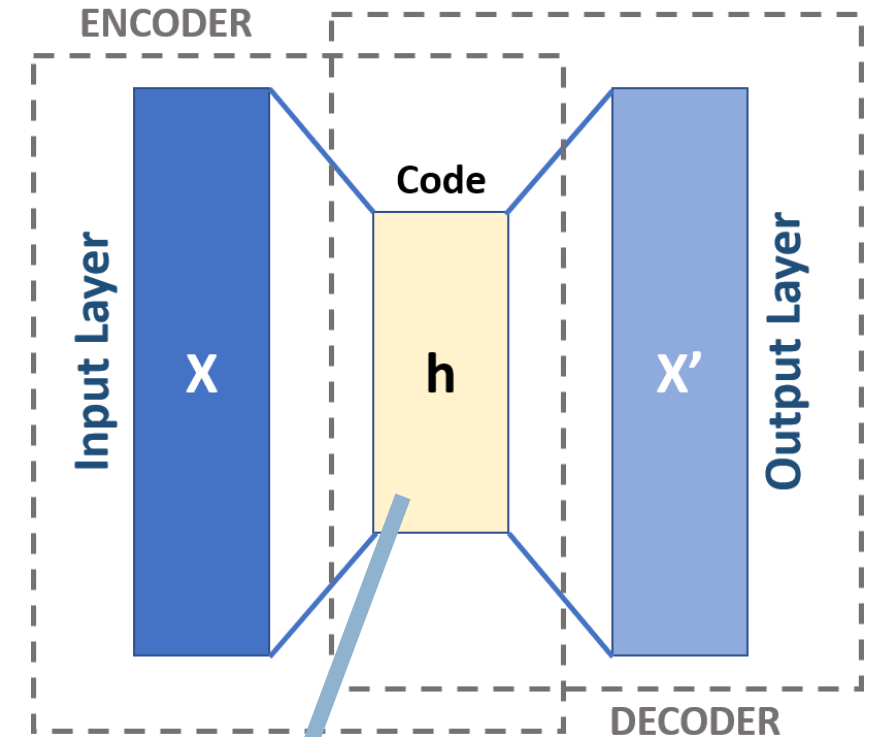
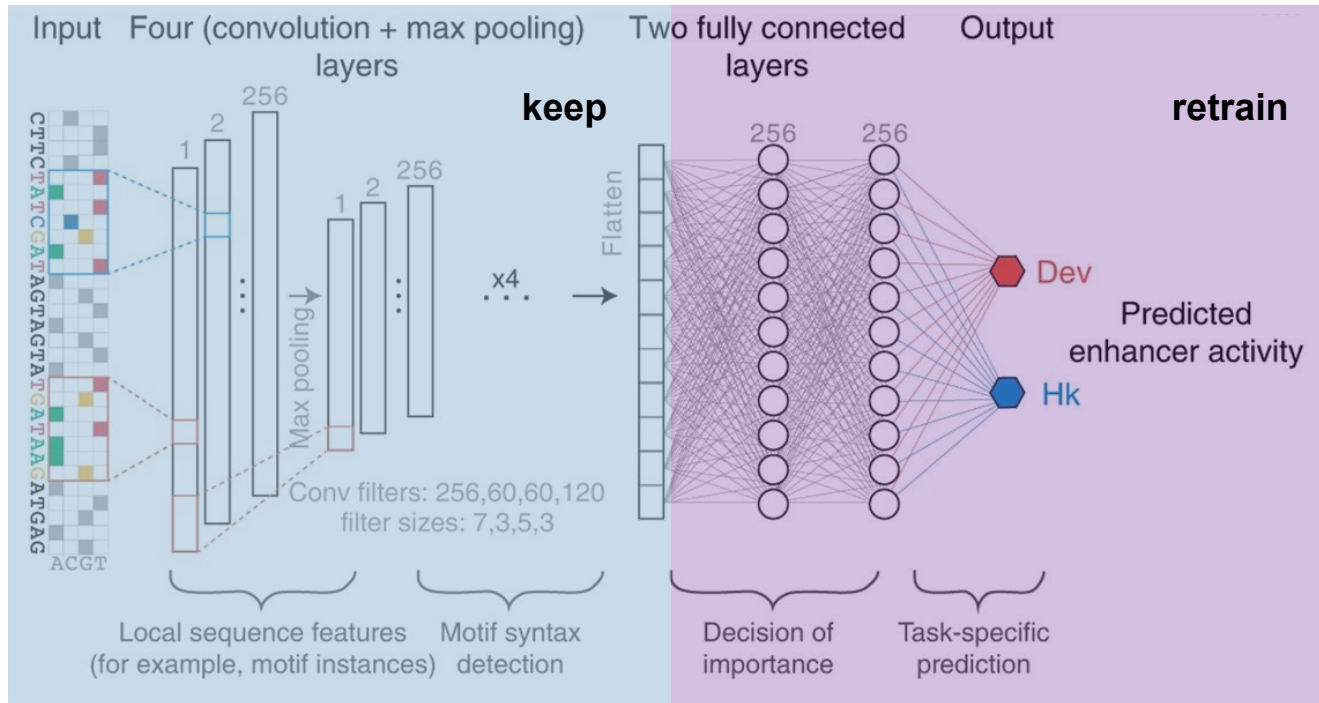
Could refer specifically to language models (e.g. DNABERT, HyenaDNA, NT) but or could use all kinds of models.





# Using pre-trained layers, data transformation from autoencoders, ...

DeepSTARR



[https://de.wikipedia.org/wiki/Autoencoder#/media/Datei:Autoencoder\\_schema.png](https://de.wikipedia.org/wiki/Autoencoder#/media/Datei:Autoencoder_schema.png)

use latent variables as input to  
(classical) machine learning models

# What is your experience or other insight?



# Feel free to reach out!

[kircherlab.github.io](https://kircherlab.github.io)

## CONTACT

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# Hands-On sessions: Model interpretation



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## Hands-on: Interpreting models with in-silico mutagenesis



[https://github.com/kircherlab/ISMB-2025\\_IGVF-MPRA-Tutorial](https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial)

**Go to:** 05\_sequence\_models/02\_ism\_and\_tfmodisco.ipynb

**Klick:** Open in Colab

# Acknowledgements

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The background of the image is a close-up, slightly blurred photograph of numerous coffee beans. The beans are dark brown with a visible crease down the middle, and they are scattered across the entire frame. The lighting is soft, creating a warm and inviting atmosphere.

**COFFEE**

**BREAK**



